

Unit – I BUSINESS INTELLIGENCE

What is business intelligence?

There is a large amount of data that is flooding businesses every day, and it is imperative that businesses can make sense of all this data. This is where BI comes into play. BI is essentially a set of technologies, architectures, and processes that help transform raw data into actionable intelligence, which can help businesses in making informed decisions.

BI directly impacts the organizational decisions of businesses. BI supports decision-making based on facts using highly calculated historical data. BI tools help in analyzing data and creating reports, dashboards, graphs, charts, and summaries.

Business Intelligence is the talk of a new changing and growing world that can be defined as a set of concepts and methodologies to improve decision-making in business through the use of facts and fact-based systems. The Goal of Business Intelligence is to improve decision-making in business ideas and analysis. Business Intelligence is not just a concept it's a group of concepts and methodologies. Business Intelligence uses analytics and gut feelings for making decisions.

Business intelligence combines business analytics, data mining, data visualization, data tools and infrastructure, and best practices to help organizations make more data-driven decisions. In practice, you know you've got modern business intelligence when you have a comprehensive view of your organization's data and use that data to drive change, eliminate inefficiencies, and quickly adapt to market or supply changes. Modern BI solutions prioritize flexible self-service analysis, governed data on trusted platforms, empowered business users, and speed to insight.

It's important to note that this is a very modern definition of BI—and BI has had a strangled history as a buzzword. Traditional Business Intelligence, capital letters and all, originally emerged in the 1960s as a system of sharing information across

organizations. The term Business Intelligence was coined in 1989, alongside computer models for decision making. These programs developed further, turning data into insights before becoming a specific offering from BI teams with IT-reliant service solutions. This article will serve as an introduction to BI and is the tip of the iceberg.

Businesses and organizations have questions and goals. To answer these questions and track performance against these goals, they gather the necessary data, analyze it, and determine which actions to take to reach their goals.

On the technical side, raw data is collected from business systems. Data is processed and then stored in data warehouses, the cloud, applications, and files. Once it's stored, users can access the data, starting the analysis process to answer business questions.

BI platforms also offer data visualization tools, which convert data into charts or graphs, as well as presenting to any key stakeholders or decision-makers.

Business intelligence refers to a collection of mathematical models and analysis methods that utilize data to produce valuable information and insight for making important decisions.

Evolution of Business Intelligence

Let us take a look at the evolution of BI over the years.

Year	Development
1999	Data remains controlled by IT departments; departments have to request reports
2003	64-bit computing is introduced; it leads to the start of in-memory processing
2004	Tableau 1.0 is released, joining QlikView among the top self-service BI tools
2006	Amazon Web Services is created by Amazon; the start of cloud computing
2007	iPhone is introduced by Apple, opening up the possibility of mobile BI; IBM buys Cognos, SAP purchases Business Objects, and Oracle acquires Hyperion
2010	Microsoft embeds Power Pivot into Excel, extending BI to ordinary business users
2012	Data visualizations help open up BI to self-service users, led by Tableau and Qlik
2015	Microsoft launches Power BI, further cementing the leading role of self-service software
2016	Augmented analytics features start a new era of BI aided by machine learning
2019	Salesforce acquires Tableau, Google buys Looker, and Sisense purchases Periscope; artificial intelligence (AI), machine learning, and natural language features become a standard
2020	Vendors expand low- or no-code, mobile BI, and multi-cloud capabilities

Components of Business Intelligence

The main components of a BI infrastructure:

Data Warehouses: These provide the source data.

Business Analytics: It helps in manipulating and mining data from the data warehouses.

Business Performance Management: It helps in monitoring and analyzing performance.

User Interfaces: These include things such as dashboards.

Main Components of Business Intelligence System are further classified as:

1. **Data Source**
2. **Data Mart / Data Warehouse**
3. **Data Exploration**
4. **Data Mining**
5. **Optimization**
6. **Decisions**

1.Data Source:

To begin, the first step is gathering and consolidating data from an array of primary and secondary sources. These sources vary in origin and format, consisting mainly of operational system data but also potentially containing unstructured documents like emails and data from external providers.

2.Data Mart / Data Warehouse:

Through the utilization of extraction and transformation tools, also known as extract, transform, load (ETL), data is acquired from various sources and saved in databases designed specifically for business intelligence analysis. These databases, commonly known as data warehouses and data marts, serve as a centralized location for the gathered data.

3.Data Exploration:

The third level of the pyramid offers essential resources for conducting a passive analysis in business intelligence. These resources include query and reporting systems, along with statistical methods. These techniques are referred to as passive because decision makers must first develop ideas or establish criteria for data extraction before utilizing analysis tools to uncover answers and confirm their initial theories. For example, a sales manager might observe a

decrease in revenues in a particular geographic region for a specific demographic of customers. In response, she could utilize extraction and visualization tools to confirm her hypothesis and then use statistical testing to validate her findings based on the data.

4.Data Mining:

The fourth level, known as active business intelligence methodologies, focuses on extracting valuable information and knowledge from data. Part II of this book will delve into various techniques such as mathematical models, pattern recognition, machine learning, and data mining. Unlike the tools discussed in the previous level, active models do not rely on decision makers to come up with hypothesis but instead aim to enhance their understanding.

5.Optimization:

As you ascend the pyramid, you'll encounter optimization models that empower you to choose the most optimal course of action among various alternatives, which can often be quite extensive or even endless. These models have also been effectively incorporated in marketing and logistics.

6.Decisions:

At last, the pinnacle of the pyramid reflects the ultimate decision made and put into action, serving as the logical end to the decision-making process. Despite the availability and effective utilization of business intelligence methodologies, the decision still lies in the hands of the decision makers, who can incorporate informal and unstructured information to fine-tune and revise the suggestions and outcomes generated by mathematical models.

Role of Business Intelligence:

The characteristics of a business intelligence analysis can be summarized by a rational and methodical approach.

- Firstly, the objectives are clearly identified and performance indicators are chosen to evaluate different options.
- Next, mathematical models are created by utilizing the connections between control variables, parameters, and evaluation metrics.
- Finally, "what-if" scenarios are explored to understand the impact of changing control variables and parameters on performance.

Process Used in Business Intelligence:

BI(Business Intelligence) uses a set of processes, technologies, and tools (such as Informatica/IBM) to transform raw data into meaningful information and then

transform information to provide knowledge. Then afterward some beneficial insights can be extracted manually and by some software then the decision-makers can make an impactful decision on the basis of insights. To sound short and clear – Business Intelligence about provides accurate information in the right and ethical format to the decision-makers of the organization. Some Important features of Business Intelligence are:

- Fact-based decision making.
- 360 degrees perspective on your business.
- Virtual team members are on the same page.
- Measurement for creating KPI (Key Performance Indicators) on the basis of historic data fed into the system.
- Identify the benchmark and then set the benchmarks for different processes.
- Business Intelligence systems can use to identify market trends and also to spot business problems that need to be identified and solved.
- Business Intelligence helps in data visualization will increase the quality of data and then also increases the quality of decision making.
- Business Intelligence systems can be used by large enterprises, and organizations along with Small and Medium Enterprises, because it is quite affordable.

Types of Business Intelligence Tools

BI tools come in a wide variety of forms. Let us discuss some of them:

- **Spreadsheets:** Spreadsheets, such as Google Docs and Microsoft Excel, are a widely used BI tool.
- **Reporting Software:** Reporting software is used to filter, organize, display, and report data.
- **Data Visualization Software:** Data visualization software are used to transform datasets into easily-readable, visually-appealing graphical illustrations to help with quick insights.
- **Data Mining Tools:** Data mining tools help mine data for recognizable patterns. This is done by using machine learning, statistics, and AI.

- **Online Analytical Processing (OLAP):** OLAP helps users analyze datasets based on various business perspectives.

Types of Users of Business Intelligence:

- **Analyst (Data Analyst or Business Analyst):** They are the statistician of the company, they used BI on the basis of historical data priorly stored in the system.
- **Head or Manager of the Company:** Head of the company uses Business Intelligence used to increase the profitability of their company by increasing the efficiency in their decisions on the basis of all the knowledge they discovered.
- **IT Engineer:** For his company.
- **Small Business Owners:** Can be used by a small businessman because it is quite affordable too.
- **Government Officials:** In the decision-making of the government.

Types of Decisions Supported by Business Intelligence:

- **Strategic Level:** The strategic level is the level where the Heads of the company decide the strategies of any business.
- **Tactical Level:** Once the strategy is made though for handling all the details and matters have a tactical level where all the technologies and methodologies come under one umbrella. This level is further responsible for continuously updating the data.
- **Operational Level:** Operation decisions are made at this level. Operational decisions help in operating the system.

Business Intelligence Architecture

The BI architecture defines the framework in which all the technologies for BI are implemented in a business. This includes IT systems and BI tools. The BI architecture is crucial in implementing a successful BI program that can use data analysis and reporting to help a business track performances, optimize processes, identify new opportunities, improve planning, and make better decisions.

Applications of Business Intelligence:

- In Decision Making of the company by decision-makers of the organizations.
- In Data Mining while extracting knowledge.
- In Operational Analytics and operational management.
- In Predictive Analytics.
- In Prescriptive Analytics.
- Making Structured data from unstructured data.
- In Decision Support System.
- In Executive Information System (EIS).

Business Intelligence Tools and Software

1. **Tableau:** A business intelligence and data visualization application that enables users to connect to different data sources, build interactive dashboards, and share findings with others.
2. **Microsoft Power BI:** A cloud-based business intelligence program that enables users to connect to a variety of data sources, produce visualizations, and communicate findings.
3. QlikView is a business intelligence and data visualization platform that enables users to build interactive dashboards and examine data in novel ways.
4. Data visualization, reporting, and analytics tools are all included in SAP BusinessObjects, a complete business intelligence suite.
5. **IBM Cognos:** A tool for performance management and corporate intelligence that enables users to build reports, scorecards, and dashboards.
6. Data visualization, reporting, and analytics technologies are all part of the full business intelligence suite known as Oracle Business Intelligence.

7. Create dynamic dashboards and reports with MicroStrategy, a business intelligence and data visualization tool.
8. Data visualization, reporting, and analytics tools are all part of the full business intelligence suite known as SAS Business Intelligence.
9. TIBCO Spotfire is a business intelligence and data visualization platform that enables users to build interactive dashboards and investigate data in novel ways.
10. **Looker:** A tool for business intelligence and data visualization that enables users to build interactive dashboards and investigate data in novel ways.

Advantages of Business Intelligence

1. Decision-making is improved because users have access to real-time data and insights through business intelligence tools. This enables users to base their decisions on correct and current information.
2. Efficiency gain: Many manual data analysis operations are automated by business intelligence systems, freeing up time and resources for other tasks.
3. Better data management: Business intelligence technologies aid in the administration and organization of data, making it simpler to locate the facts required for decision-making.
4. Greater visibility: Business intelligence solutions give users a comprehensive picture of the functioning of the firm, enabling them to spot areas that could use improvement.
5. A better understanding of customers: Business intelligence technologies helps firms understand their customers better, enabling them to customize products and services to suit their needs.
6. Cost savings: Business intelligence technologies assist firms in locating inefficiencies and cost savings, which boosts revenue.
7. Better forecasting: Organizations may evaluate past data and predict future patterns using business intelligence technologies, which enables them to plan more successfully for the future.
8. Competitive advantage: By granting access to important data and insights that can guide them in making better decisions, business intelligence technologies provide firms a leg up on their rivals.

9. Collaboration is improved as a result of using business intelligence technologies to disseminate information between teams and departments. This promotes better decision-making and collaboration.
10. Better Monitoring: Business intelligence technologies assist firms in tracking important metrics like revenue, customer happiness, and staff performance and in monitoring performance.
11. Boosts Productivity - Businesses can create reports with just one click using BI programs. This saves a lot of time and resources. This helps employees in being more productive in their tasks.
12. Improves Visibility - BI makes sure that all processes are more visible and any areas that need attention can be focused on.
13. Fixes Accountability - BI systems help set accountability in businesses as there should be someone accountable for the businesses' performance against the set goals.
14. Gives a Bird's Eye View - BI systems provide a bird's eye view through BI features, dashboards, and reports, which helps decision-makers.
15. Streamlines Business Processes - BI systems help remove complexities from business processes. BI systems automate analytics, offer predictive analysis and benchmarking, etc.
16. Allows for Easy Analytics - Even nontechnical professionals and non-analysts can collect and process data quickly using BI systems.

Disadvantages of Business Intelligence

1. **Complexity:** The implementation and upkeep of business intelligence systems can be extremely difficult and complicated. This may be a drawback for companies with constrained IT resources.
2. **High costs:** Some businesses find it prohibitively expensive to implement and purchase business intelligence technologies.
3. Business intelligence strongly depends on accurate and current data. The insights produced by business intelligence technologies could not be accurate if the data is inconsistent, erroneous, or incomplete.
4. **Data Security:** Business intelligence systems handle and store a lot of sensitive data, which, if not adequately protected, is susceptible to security breaches.
5. **Dependence on IT:** Because business intelligence solutions frequently rely largely on IT assistance, it may be challenging for enterprises to quickly get the data they require.

6. **Limited scalability:** For firms with huge data volumes, business intelligence solutions may not be able to handle enormous amounts of data.

BI Methods

Much more than a specific “thing,” business intelligence is an umbrella term that covers the processes and methods of collecting, storing, and analyzing data from business operations or activities to optimize performance. All of these things come together to create a comprehensive view of a business to help people make better, actionable decisions. Over the past few years, business intelligence has evolved to include more processes and activities to help improve performance. These processes include:

- **Data mining:** Using databases, statistics, and machine learning (ML) to uncover trends in large datasets
- **Reporting:** Sharing data analysis to stakeholders so they can draw conclusions and make decisions
- **Performance metrics and benchmarking:** Comparing current performance data to historical data to track performance against goals, typically using customized dashboards
- **Descriptive analytics:** Using preliminary data analysis to find out what happened
- **Querying:** Asking the data-specific questions, BI pulling the answers from the data sets
- **Statistical analysis:** Taking the results from descriptive analytics and further exploring the data using statistics such as how this trend happened and why
- **Data visualization:** Turning data analysis into visual representations such as charts, graphs, and histograms to more easily consume data
- **Visual analysis:** Exploring data through visual storytelling to communicate insights on the fly and stay in the flow of analysis
- **Data preparation:** Compiling multiple data sources, identifying the dimensions and measurements, and preparing it for data analysis

How BI, data analytics, and business analytics work together

Business intelligence includes data analytics and business analytics but uses them only as parts of the whole process. BI helps users draw conclusions from data analysis. Data scientists dig into the specifics of data, using advanced statistics and predictive analytics to discover patterns and forecast future patterns.

Data analytics asks, “Why did this happen and what can happen next?” Business intelligence takes those models and algorithms and breaks the results down into actionable language. According to Gartner's IT glossary, “business analytics includes data mining, predictive analytics, applied analytics, and statistics.” In short, organizations conduct business analytics as part of their larger business intelligence strategy.

BI is designed to answer specific queries and provide at-a-glance analysis for decisions or planning. However, companies can use the processes of analytics to continually improve follow-up questions and iteration. Business analytics shouldn't be a linear process because answering one question will likely lead to follow-up questions and iteration. Rather, think of the process as a cycle of data access, discovery, exploration, and information sharing. This is called the cycle of analytics, a modern term explaining how businesses use analytics to react to changing questions and expectations.

Difference between Information and Intelligence

In today's fast-paced world, we are flooded with information from all directions. Social media, news outlets, and even our devices bombard us with endless data streams. However, have you ever wondered if all this information is helping us make better decisions? The answer may lie in the distinction between information and intelligence.

However, these two concepts aren't the same, and understanding their differences is crucial for individuals, organizations, and societies to make informed decisions, anticipate changes, and mitigate risks. Understanding the nuances between the two can profoundly impact our ability to interpret and utilize the data we encounter daily.

What is Information?

Information is data or facts collected or received about a particular subject. It can be objective or subjective, accurate or inaccurate, complete or incomplete, and relevant or irrelevant. Examples of information include news articles, financial statements, customer feedback, social media posts, and scientific research. Information can be received from various sources, such as books, websites, surveys, interviews, and sensors.

The characteristics of information can vary depending on the context, purpose, and audience. For instance, information intended for a general audience may be presented in a simplified and engaging format. In contrast, information that is intended for a specialized audience may be presented in a technical and precise format. However, some common characteristics of information include:

- **Verifiability:** Information can be validated through evidence, sources, or documentation.
- **Relevance:** Information is related to the subject matter or question at hand.
- **Timeliness:** Information is current or up-to-date.
- **Accessibility:** Information is available and accessible to those who need it.
- **Diversity:** Information comes from different sources and perspectives.

The sources of information can also vary depending on the type and quality of the information. For instance, one source of information based on scientific research may be more reliable than information based on personal anecdotes. Some familiar sources of information include:

- **Primary sources:** Information from the subject matter, such as data, statistics, and surveys.
- **Secondary sources:** Information from published sources, such as books, articles, and reports.
- **Tertiary sources:** Information from compilations or summaries of primary and secondary sources like encyclopaedia's, handbooks, and databases.

What is Intelligence?

Intelligence refers to knowledge or insights from information through analysis, synthesis, and interpretation. It involves understanding the information's

relationships, patterns, trends, and implications to make predictions, judgments, and recommendations.

Intelligence examples include market research reports, threat assessments, risk analyses, performance metrics, and strategic plans. Various methods, such as data mining, machine learning, expert judgment, and scenario planning, can generate intelligence.

Intelligence's characteristics differ from information's because intelligence involves more intelligence analysis, synthesis, and interpretation. Some common characteristics of intelligence include:

- **Insightful:** Intelligence provides a deeper understanding and explanation of the subject matter.
- **Predictive:** Intelligence forecasts future trends and outcomes based on current data and analysis.
- **Strategic:** Intelligence supports decision-making and planning at a higher level of abstraction.
- **Actionable:** Intelligence provides recommendations or courses of action based on the analysis and interpretation.

The intelligence sources can also vary depending on the complexity and scope of the intelligence process analysis. For instance, intelligence intended for a specific decision or action may require more expertise and resources than intelligence intended for general awareness. Some common sources of intelligence include:

- **Human expertise:** Intelligence generated by subject-matter experts with knowledge and experience in a particular field.
- **Machine intelligence** is intelligence generated by algorithms, models, and simulations that analyze and interpret data.
- **Collaborative intelligence:** Intelligence that is generated by a group of people who share knowledge and insights to solve a complex problem or challenge.

Differences between Information and Intelligence

Although both information and intelligence are related, some fundamental differences set them apart. Here are the main differences between both information and intelligence:

1. Level of Analysis

Information is often raw and requires further analysis to extract valuable insights and understanding. The analysis process involves organizing, filtering, and interpreting the data to make it worthwhile.

On the other hand, intelligence involves a higher level of analysis, synthesis, and interpretation of the data. It requires subject matter expertise and a first sense of deep understanding of the data to extract insights and understanding that can be used for decision-making.

2. Purpose

The purpose of the information is to provide data and knowledge about a particular subject. It can be used to answer questions or provide context for decision-making. On the other hand, intelligence is designed to provide insights and understanding derived from the information that can be used for decision-making and planning.

Intelligence aims to provide insights that can be used to gain a competitive advantage, identify opportunities, and make informed business decisions.

3. Time Horizon

Information often focuses on the present or past, while intelligence focuses on the future. Intelligence seeks to anticipate changes, risks, and opportunities based on analysis and interpretation of the data.

4. Level of Abstraction

Information is often presented at a lower level of abstraction, such as raw data, facts, or events. Intelligence is presented at a higher level of abstraction, such as patterns, trends, and implications.

5. Sources

The sources of information are often diverse and can be obtained from primary, secondary, or tertiary sources. Information can come from various sources, including internal and external sources. It can be structured or unstructured and in different formats, such as text, images, or audio.

The intelligence sources are often more specialized and require more expertise and resources, such as human intelligence, machine intelligence, or collaborative intelligence. Conversely, intelligence is derived from the information through analysis, synthesis, and interpretation. Intelligence is typically generated by subject matter experts or analysts who have a deep understanding of the subject matter.

6. Process

Generating intelligence involves several steps, such as collecting, analyzing, synthesizing insights, and presenting recommendations. Generating information may involve some of these steps but not all of them.

Why Does Understanding the Difference Matter?

Understanding the difference between information and intelligence is crucial for several reasons:

1. Effective decision-making

Effective decision-making requires intelligence, not just information. Intelligence provides insights and understanding derived from information that can be used for decision-making and planning. By understanding the stark difference between information and intelligence, decision-makers can ensure they have the correct data and insights to make informed decisions.

2. Competitive Advantage

Organizations that extract meaningful insights from data are more likely to gain a competitive advantage in today's data-driven world. By understanding the difference between information and intelligence, organizations can ensure that they are generating and leveraging intelligence to gain insights that can be used to make informed decisions and identify new opportunities.

3. Resource Allocation

Understanding the difference between access to information and intelligence can help organizations allocate resources more effectively. Organizations that generate intelligence instead of just collecting information can focus their resources on analyzing and interpreting data, leading to more valuable insights.

4. Innovation

Intelligence can be used to identify new opportunities and potential innovations. By understanding the difference in value between information and intelligence, organizations can generate the insights they need to identify new trends, patterns, and relationships that can be leveraged to drive innovation.

5. Risk Management

Intelligence can also be used to manage risk. By analyzing and interpreting data, organizations can identify potential risks from threat actors and take proactive measures to alleviate them. By understanding the difference between information and intelligence, organizations can generate insights into threat actors, which they need to manage risk effectively.

6. Improved Communication

Understanding the difference between information and intelligence can also improve organizational communication. By clearly defining the terms, objectives, and concepts related to data, decision-makers, and analysts can communicate more effectively, reducing the risk of misinterpretation and confusion.

In conclusion, information and intelligence are two related but distinct concepts that play a critical role in decision-making, planning, and innovation. Information is raw data or facts collected or received about a particular subject. At the same time, intelligence is knowledge or insights derived from information through analysis, synthesis, and interpretation.

Understanding the differences between information and intelligence is crucial for individuals, organizations, and societies to make informed decisions, anticipate changes, and mitigate risks. By recognizing the unique characteristics and sources of information and intelligence, decision-makers can ensure they have full access to the insights and understanding they need to navigate an increasingly complex and data-driven world.

Difference Between Artificial Intelligence & Human Intelligence

Artificial Intelligence:

Artificial Intelligence is based on human insights that can be decided in a way that a machine can effortlessly actualize the tasks, from the basic to those that

are indeed more complex. The reason for manufactured insights is learning, problem-solving, reasoning, and perception.

This term may be connected to any machines which show related to a human intellect such as examination and decision-making and increments the efficiency.

AI covers assignments like robotics, control systems, face recognition, scheduling, data mining, and numerous others.

Advantages of Artificial Intelligence (AI):

- AI can process vast amounts of data much faster than humans.
- AI can work around the clock without needing breaks or rest.
- AI can perform tasks that are too dangerous or difficult for humans.

Disadvantages of Artificial Intelligence (AI):

- AI lacks the creativity and intuition that humans possess.
- AI is limited by its programming and may not be able to adapt to new or unexpected situations.
- AI may make errors if not programmed and trained properly.

Human Intelligence:

Human intelligence or the behavior of the human being has come from past experiences and the doings based upon situation, and environment. And it is completely based upon the ability to change his/her surroundings through knowledge which we gained.

It gives diverse sorts of information. It can provide data on things related to a particular aptitude and knowledge, which can be another human subject, or, within the case of locators and spies, diplomatic data which they had to get to. So, after concluding all it can give data on interpersonal connections and arrange of interest.

Advantages of Human Intelligence (HI):

- HI has creativity, intuition, and emotional intelligence that AI lacks.
- HI can adapt to new and unexpected situations.
- HI can provide ethical and moral considerations in decision-making.

Disadvantages of Human Intelligence (HI):

- HI is limited by its physical and mental capabilities.

- HI is prone to biases and may make errors or poor decisions.
- HI requires rest and breaks, which can slow down processes.

Similarities between Artificial Intelligence (AI) and Human Intelligence (HI):

- Both AI and HI can learn and improve over time.
- Both AI and HI can be used to solve complex problems and make decisions.
- Both AI and HI can process and interpret information from the world around them.

Below is a table of differences between Artificial intelligence and Human intelligence:

No.	Feature	Artificial Intelligence	Human Intelligence
1.	Emergence	AI is an advancement made by human insights; its early improvement is credited to Norbert Wiener who theorized on criticism mechanisms.	On the other hand, human creatures are made with the intrinsic capacity to think, reason, review, etc.
2.	Nature	Artificial intelligence (AI) strives to build machines that can mimic human behavior and carry out human-like tasks.	Human intelligence seeks to adapt to new situations by combining a variety of cognitive processes.
3.	State	Machines are digital.	The human brain is analogous.
4.	Function	AI-powered machines rely on input of data and instructions.	Humans use their brains' memory, processing power, and cognitive abilities.

No.	Feature	Artificial Intelligence	Human Intelligence
5.	Pace/Rate of AI and human	As compared to people, computers can handle more data at a speedier rate. For occurrence, in the event that the human intellect can solve a math problem in 5 minutes, AI can solve 10 problems in a minute.	In terms of speed, humans cannot beat the speed of AI or machines.
6.	Learning ability	As machines are unable to reason abstractly or draw conclusions from the past. They can only acquire knowledge through information and frequent training, but they will never develop a human-specific thinking process.	Learning from various events and prior experiences is the foundation of human intelligence.
7.	Decision Making	AI is profoundly objective in choice making because it analyzes based on absolutely accumulated data.	Human choices may be affected by subjective components which are not based on figures alone.
8.	Perfection	AI frequently produces precise comes about because its capacities are based on a set of modified rules.	For human insights, there's more often than not room for "human error" as certain subtle elements may be missed at one point or the other.

No.	Feature	Artificial Intelligence	Human Intelligence
9.	Energy Consumption	The modern computer generally uses 2 watts of energy.	On the other hand, human brains uses about 25 watts
10.	Modification of AI and Human	AI takes much more time to adjust to unused changes.	Human insights can be adaptable in reaction to the changes in their environment. This makes individuals able to memorize and ace different skills.
11.	Versatility	AI can as it were perform fewer assignments at the same time as a framework can as it were learn duties one at a time.	The human judgment skills underpin multitasking as proven by differing and concurrent roles.
12.	Social Networking	AI has not aced the capacity to choose up on related social and enthusiastic cues.	On the other hand, as social creatures, people are much way better at social interaction since they can prepare theoretical data, have self-awareness, and are delicate to others' feelings.
13.	Task	It does optimization of the system. It cannot be creative or innovative as humans can only think and machines cannot.	It is innovative or creative.

No.	Feature	Artificial Intelligence	Human Intelligence
14	Processing	Based on algorithms and mathematical models	Based on cognitive processes and biological structures
15	Learning	Based on data and feedback loops	Based on experience, intuition, and creativity
16	Speed	Can process data and perform tasks much faster than humans	Slower than AI in processing large amounts of data, but can make complex decisions quickly
17	Adaptability	Can quickly adapt to new data and situations	Can adapt to new situations, learn from experience, and make decisions based on context
18	Emotions	Lacks emotions and empathy	Capable of feeling emotions and empathy
19	Creativity	Limited ability to be creative or think outside of the box	Capable of creativity, imagination, and innovation
20	Ethics	Does not have a moral code or conscience	Has a moral code and conscience that guides

No.	Feature	Artificial Intelligence	Human Intelligence
21	Physical Limitations	Does not have physical limitations, can operate 24/7	decision-making Limited by physical capabilities and requires rest and maintenance

Artificial Intelligence (AI) mimics specific cognitive abilities but lacks the depth of the human mind. Human intelligence encompasses cognitive abilities, emotions, creativity, and adaptability. The true potential lies in synergizing AI's data processing prowess with human intuition, ethics, and contextual understanding to augment capabilities while aligning with societal values.

Business Intelligence Value Chain

What is business intelligence value chain?

Business Intelligence (BI) is a type of architecture that supports collecting and analysing data to come up with meaningful information about the business. Owing to the competitive environment, companies are heavily investing in data warehousing, analytics, and reporting for informed decision-making. This whole decision-making process comprises a chain of methodologies, and is known as the 'Business Intelligence Value Chain'.

This system is designed to have a methodological structure to take raw data and generate meaningful insights from them that leverage business value. Over time, this system has evolved to have more concrete and properly defined stages. Leaders can use the BI value chain to form better strategies, processes, structures, and systems to deal with internal and external threats.

5 Stages of BI Value Chain

There are several stages in converting raw data into meaningful information that, in turn, adds value to the business. These stages are typically referred to as the stages of a Business Intelligence Value Chain. These stages are the following,

1. Data sourcing

The first step in the BI process is accessing the data. This stage deals with the storage, management, and accessing of raw data. Thus, data warehousing becomes an essential aspect of business intelligence. Sourcing data is of utmost importance as the quality and relevancy of data directly impact the quality of insights gained and the consequent decision taken by the leadership. Another aspect of data sourcing deals with identifying and exploring various data resources. It's crucial for the following reasons-

- Data can be related to each other
- It can be unstructured (in the form of text, images, or other forms of unstructured data)
- Data can have peculiar features (such as different data types etc.)

It is vital for a business intelligence analyst to know what kind of data is available and where and how they can access it.

2. Data engineering and analysis

After data sourcing, the next logical step is information engineering and analysis. To perform analysis on the data and to gain even the fundamental insights from the data, a BI analyst needs to have the data in a structure that is conducive for analysis. This requires data engineering that includes-

- Converting data into a structured (tabular) format
- Removing or imputing missing values
- Capping outliers
- Removing multicollinearity

Once the data is structured and ready, **Exploratory Data Analysis (EDA)** is performed through which useful information can be synthesized, such as-

- Summarizing data using descriptive statistics
- Finding associations in data

After EDA, Structured Data Analysis (SDA) is performed, which includes three types of analysis-

1. **Trend Analysis:** It's used for identifying patterns in the data. This includes, for example, assessing sales based on different geographical regions, the volume of product sales over a stipulated period, etc.

2. **Mathematical Analysis:** It's used for calculating performance and growth using mathematics. This includes, for example, assessing the margin of sales and growth in absolute and in percentage.
3. **Statistical Analysis:** Inferential statistics is used to assess the statistical significance of the patterns and peculiarities being identified in the data. Statistics is also used to perform predictions, analytics, and forecasting using regression and other methods. The statistical coefficients can also be used to perform prescriptive analytics, identifying reasons for certain business phenomena. Therefore, the model building can take place at this stage of Business Intelligence.

To understand how data is analysed, you must also learn about the univariate analysis of data.

3. Situation awareness

The core of business intelligence is to provide concerned individuals in an organization with knowledge helping in situational awareness. This stage in Business Intelligence deals with report creation and presentations that provide the decision-makers with essential and relevant information to help them be aware of the events in and around the organization. This information can make them aware of, for example, government policies, upcoming market trends, market forces, etc.

4. Decision-making

Once the decision-makers know the 'what and why' of the events in and around the business, the next stage is proactive decision-making and its evaluation. The insights, knowledge, and intelligence gained from analytics enable decision-makers to make data-driven decisions. This can be, for example, coming up with strategic decisions such as management change, management of products and categories, opening or shutting of offices, launching of new products, or can be operational, such as promotional campaigns, upsell and cross-sell campaigns, etc.

5. Decision support

This stage deals with supporting the proposed decision by evaluating it. Evaluation includes identifying the risks, opportunities, benefits, profit, pitfalls, cost-to-benefit ratio, return on investment, and estimating the business value of the proposed decisions. All of this helps in making efficient and effective decision-making.

Another way of setting the stages of a Business Intelligence Value chain is to **lay them out in terms of complexity**. In terms of complexity, the stages are-

- **Reporting:** The most basic stage of BI is to understand what is happening and report the same.
- **Analysis:** The next stage is to answer 'why did it happen?' and identify the cause and effect of the events in and around the business.
- **Monitoring:** This stage deals with answering the question—' what is happening right now?' as this deals with keeping an eye on the business's day-to-day activities.
- **Predicting & Forecasting:** The next stage is significantly more complex than the previous one as this involves using predictive statistics and even machine learning. Here patterns and trends are assessed through a predictive or forecasting model. This helps answer questions like 'what will happen given the past data?'
- **Prescriptive Modelling:** The most complex stage involves prescriptive models to answer questions like 'What will happen, when will it happen, and why will it happen?'. All this requires understanding complex relationships and interactions between the data features. This stage also deals with re-prediction and re-forecasting to create a feedback loop that helps make models better.

Understanding the value of BI

Business intelligence is essential for any business. Here's how BI enables a business:

1. **Cost reduction:** The most significant value of Business Intelligence is that it can significantly reduce costs for an organization. This can be achieved by increasing operational efficiency, identifying and eliminating backlogs and delays, performing root cause identification, looking for a solution, and eliminating wasteful resources.
2. **Increase in revenue:** BI's next significant value addition is helping an organization increase its revenue. Information sharing among leadership and end-users, better market analysis, upskilling the existing workforce, and analysing customer behaviour to design products can help increase revenue.
3. **Enhancing customer satisfaction:** For any business, increasing the satisfaction of its end-user is of paramount importance. While

customer satisfaction can be achieved in many ways, BI can help increase customer satisfaction by providing customers with factual information, performing competitive analysis to understand the pros and cons of the product offered by competitors, and analyzing the issues faced by the customers to help in a quick resolution.

Several factors of a business intelligence system influence it and can make or break it. While there are hundreds of factors affecting a business intelligence system, we can group them in the following manner.

Factors influencing the BI system

1. Management:

Top management is essential in BI as it provides the directives and necessary support. A team of well-qualified managers is required as their active participation is crucial for the success of any BI project.

2. Business Understanding:

Another essential factor of a business intelligence system is how well it links to the business and solves a business problem. A BI project must be well defined, have a straightforward methodology to manage the project, should be managed by the technical as well as the business side, and should have clear goals.

3. Project Manager:

A BI project must be divided into achievable chunks or sub-projects that achieve the larger objective. Each of these projects can have analytical and technical individuals, but the project manager must have an excellent understanding to make sense of the insights provided by the team.

4. Strategy:

An essential factor in a BI project's success is clear business vision and planning. This can be achieved by spending extensive time creating the strategic framework.

5. Change in Management:

The high-level leadership must be flexible with the team managing the BI project. If the need arises to change the management, it must be done.

6. Project Monitoring:

The project performance must to monitored regularly. The solution provided should be as per the requirement of the end-user, and the project should be flexible as per the changing requirement of the end-user.

7. Human Talent:

The teams involved in the BI project, from the BI project manager to all the team members, should be competent with a clear understanding of the business link and goal. A good balance and mix of skill sets in the team can considerably improve the performance

8. Learning & Upskilling:

Given the dynamic nature of technology, the resources must be upskilled in the newest BI tools and methodologies.

9. Environment:

The environment in which BI analysts and related individuals work should be such that respective people solve technical and non-technical issues on time.

10. Scalability:

The BI products developed in the project must be modular as this help in scalability.

Business Intelligence Lifecycle

A business intelligence life cycle is comprised of several steps.

Few BI lifecycle steps coincide with the 5 stages of BI value chain as well.

A typical business intelligence life cycle has the following steps-

1. Identifying business process and problem

The first step in a business intelligence life cycle is to identify the business problem that BI will aim to solve. This involves understanding the business process, identifying where it is lacking, and the problem being raised from there. For example, the business process of acquiring a new customer through digital marketing can have the issue of budget constraints, causing a lack of new acquisitions. Thus, the problem becomes a lack of new customers, and the question that BI needs to solve now is “How to get new customers efficiently”.

2. Data collection

The next step is to look for data. Continuing the above example, a BI analyst has to look for relevant data that explain how much money was spent on different streams of marketing, the number of customers acquired from each stream, the performance of customers belonging to different streams, customer complaints, and behaviour, purchasing history, etc. All this data can be available at various places in different forms. You can use data collection tools to compile important data easily.

3. Data warehousing

Another significant aspect of a business intelligence life cycle is data warehousing. This step involves saving relevant data in RDBMS and other types of database management systems.

4. Data preparation and analytics

The fourth step is to prepare the data stored in a warehouse to perform analytics. This includes monitoring daily activities, coming up with summary stats, and different models such as predictive, prescriptive, forecasting, etc.

5. Reporting

Reporting involves converting the insights gained from analytics into visual friendly and 'digestible' forms of reports, dashboards, KPIs, and presentations provided to the decision-makers.

6. Business decision

Based on the reports, the leadership takes strategic and operational decisions. As data back this decision-making, it has a high probability of successfully achieving its objectives. For example, a report can explain the daily acquisition of customers from different marketing streams, the statistical impact of marketing campaigns on acquisitions, and the prediction of the acquisition's magnitude given a revised amount for marketing. Based on the report, the leadership may decide to cut the funding for digital marketing through Facebook and reallocate the money to Instagram.

7. Evaluation and iteration

The last step is to assess the feasibility of the decision in monetary or even social terms. Once the decision is implemented, it's essential to check if the decision was successful and to what extent. This feedback is critical as specific tweaks can be made in the decision to make it more effective, thus making the process iterative.

Now let's take a look at how BI functions in logistics and transportation.

BI in Logistics and Transportation

Organizations across all domains understand the value of BI- from finance to NGOs. However, over the last few decades, the significance of business intelligence in logistics & transportation has drastically changed this industry for good. Today, many companies outsource their logistic work to third party logistics providers (3PLP). These companies have heavily involved business intelligence in their work and have resulted in performance improvement of their services. However, BI also helps streamline the organizational and IT aspects of such companies.

The BI value is evident in the services by 3PLP. Through detailed monitoring of the daily working and in-depth analytical reports on the functioning of these services, these companies have made it possible to improve and become profitable.

Transportation Management

BI helps in route optimization, traffic control, efficient selection of mode of transportation, etc. Various types of analysis are performed here, such as

- Carrier performance evaluation to select the **best third-party carrier**
- Analysis of the **performance of truck drivers to maximize performance with safety**
- Capacity analysis to **better manage the resources and minimize loss of revenue by planning** capacity in advance
- Analysis of supplier compliance to find **leakages in payments** and reduce delays
- **Scheduling route that takes minimum time and fuel**
- Analysis of **mode-cost to identify the best mode of transportation** for different products

Warehouse & Inventory Management

Many logistics companies store produce they transport, and managing the warehouse becomes essential. Here BI helps in-

- **Analysis of inventory to understand the currently available products and their associated information (shape, size, and shelf life).**
 - Analysis of the performance of warehouses by creating KPIs (such as on-time shipments, time taken to pick up products from warehouses, etc.)
- 

- Analysis of warehouse costs by analyzing the dimensions and handling requirements of the products stored there.
- Analysis of products picked from warehouses to efficiently group products and place them around each other during storage
- Analysis of warehouse space utilization to better understand how the space is being utilized to use it efficiently

Value Added Service

3PLP also provides other services such as handling reverse logistics, packing, labeling, assembly, etc. These value-added services are the ones that often make companies stand out. However, business intelligence in logistics is used to identify the cost-benefit analysis of these services and analyze each of these services to manage their workings efficiently.

As mentioned earlier, BI not only helps the logistics companies make their services better but also helps provide organizational and IT support. For example, BI provides organizational support through better financial, human resources, marketing, sales, and corporate management. On the other hand, BI delivers IT support to such companies by improving supply chain visibility, performing forecasting, and customer reports. Going into the details of all such aspects is beyond the scope of this article, but you certainly can explore them.

Below are a few commonly asked questions on the Business intelligence value chain. If you have more queries, drop in the comment section or reach out to us directly.

The 4 elements of a successful business intelligence environment

In increasingly complex data systems, ensuring that data is accurate, consistent, and timely can be a daunting task. Fortunately, by understanding the four elements of a business intelligence (BI) environment, companies can develop a BI strategy that effectively meets their needs.

But why utilize business intelligence in the first place?

In a nutshell, business intelligence allows organizations to make data-driven decisions. This process begins with data gathering, followed by data cleaning and standardization, analysis, and reporting.

All four of these steps are essential to creating an effective BI strategy. By learning more about each element, companies can create a BI plan that improves their bottom line.

In this article, we will discuss the four elements of a business intelligence environment: data, people, processes, and technology, and how they impact business intelligence strategy.

- What are the four environmental factors that can impact a business intelligence strategy?
- How do each of these four factors influence data gathering, data cleaning/standardization, analysis, and reporting?
- How can companies ensure that their business intelligence environment is effective and efficient?
- What are some common mistakes that companies make when it comes to their business intelligence environment?

What are the four environmental factors that can impact a business intelligence strategy?

In order to understand how a holistic business intelligence strategy is possible, it's first necessary to understand the four environmental factors that can impact such a strategy.

The four areas can be broken down into two main categories: internal and external. Internal factors are those that the company has direct control over, while external factors are out of the company's control but can still impact the business intelligence strategy.

The four environmental factors are:

1. Data: This is the most important factor in business intelligence, as without data there is nothing to analyze or report on. Data can come from a variety of sources, both internal and external to the organization.

2. People: The people involved in business intelligence play a critical role in its success. From the data analysts who gather and clean the data, to the business users who interpret and use the data to make decisions, each person involved must have a clear understanding of their role in the process.

3. Processes: The processes used to gather, clean, and analyze data must be well-designed and efficient in order to produce accurate and timely results.

4. Technology: The technology used to support business intelligence must be up to date and able to handle the volume and complexity of data.

Now that we've broken down the four environmental factors, let's take a closer look at how each of these factors can influence business intelligence strategy.

How do each of these four factors influence data gathering, data cleaning/standardization, analysis, and reporting?

1. Data

Business intelligence thrives on data. Without data, there is nothing to analyze or report on. Data can come from a variety of sources, both internal and external to the organization.

Internal data sources can include things like transaction data, customer data, financial data, and operational data. External data sources can include public records, social media data, market research data, and competitor data.

The data gathering process must be designed to collect the right data from the right sources. Once the data is gathered, it must then be cleaned and standardized so that it can be properly analyzed.

In the BI environment, data is king. All other factors must be aligned in order to support the data and help it reach its full potential.

2. People

The people involved in business intelligence play a critical role in its success. From the data analysts who gather and clean the data, to the business users who interpret and use the data to make decisions, each person involved must have a clear understanding of their role in the process.

The data analysts need to be able to collect data from all relevant sources, clean and standardize the data, and then load it into the BI system. The business users need to be able to access the data, understand what it means, and use it to make decisions.

This can take many forms, but a key component is data literacy: the ability to read, work with, analyze, and argue with data. Data literacy is essential for business users to be able to make decisions based on data.

In a successful business intelligence environment, people are trained and empowered to use data to make decisions.

3. Processes

The processes used to gather, clean, and analyze data must be well-designed and efficient in order to produce accurate and timely results.

The data gathering process must be designed to collect the right data from the right sources. Once the data is gathered, it must then be cleaned and standardized so that it can be properly analyzed.

The data analysis process must be designed to answer the right questions. The results of the data analysis must be presented in a way that is easy to understand and use.

4. Technologies

The technology used to support business intelligence must be up to date and able to handle the volume and complexity of data.

The BI system must be able to collect data from all relevant sources, clean and standardize the data, and then load it into the system. The system must be able to support the data analysis process and provide easy-to-use tools for business users to access and analyze the data.

You want your BI technology to offer features such as self-service analytics, predictive analytics, and social media integration. However, the technology must be easy enough to use that business users don't need a Ph.D. to use it.

In a successful business intelligence environment, the technology is easy to use and provides the features and functionality needed to support the data gathering, analysis, and reporting process.

How can companies ensure that their business intelligence environment is effective and efficient?

Without the proper attention, a business intelligence environment can quickly become ineffective and inefficient. The best way to avoid this is to have a clear understanding of the four environmental factors that impact business intelligence.

For companies to be successful with business intelligence, they must understand how to gather data from all relevant sources, how to clean and standardize the data, how to load it into the BI system, and how to use it to make decisions.

In addition, companies need to ensure that their people are trained and empowered to use data to make decisions. And finally, companies need to have the right technology in place to support their business intelligence efforts.

For example, a company may have the best data gathering and analysis processes in place, but if the business users don't understand how to use the data, then the efforts will be for naught.

Or a company may have the right technology in place, but if the data isn't being collected properly, then the technology won't be able to do its job.

It is only by taking into account all four of these environmental factors that companies can ensure that their business intelligence environment is effective and efficient. Then, and only then, can they reap the benefits of business intelligence.

What are some common mistakes that companies make when it comes to their business intelligence environment?

While business intelligence can be extremely beneficial to companies, there are some common mistakes that companies make when it comes to their business intelligence environment.

1. Not understanding the different types of data

There are three different types of data that need to be considered when gathering data for business intelligence:

Structured data is data that is organized in a predefined format. This type of data is typically stored in a database and can be easily accessed and analyzed.

Unstructured data is data that is not organized in a predefined format. This type of data includes things like emails, social media posts, and images.

Semi-structured data is data that is partially structured and partially unstructured. This type of data includes things like XML files and JSON files.

Companies need to understand the different types of data and how to best gather and analyze each type.

2. Not gathering data from all relevant sources

In order to make accurate decisions, companies need to gather data from all relevant sources. This includes internal data sources like financial reports and customer databases as well as external data sources like news articles and social media posts.

3. Not cleaning or standardizing the data

Once the data has been gathered, it needs to be cleaned and standardized. This includes things like removing duplicates, standardizing formats, and filling in missing values.

4. Not loading the data into the BI system effectively

Once the data has been cleaned and standardized, it needs to be loaded into the BI system. This can be done manually or through an automated process.

5. Not analyzing the data properly

Once the data is in the BI system, it needs to be analyzed. This includes things like creating reports, running queries, and building dashboards.

6. Not making decisions related to the data

Once the data has been analyzed, it needs to be used to make decisions. This includes things like setting prices, launching new products, and hiring new employees.

7. Not empowering people to use data

In order for business intelligence to be successful, people need to be empowered to use data to make decisions. This includes training people on how to use the BI system, giving them access to the data, and encouraging them to use data when making decisions.

As a Conclusion, Business intelligence can be extremely beneficial to companies, but only if it is done correctly.

In order to make sure that your business intelligence environment is effective, you need to take into account the four environmental factors: data, people, processes, and technology.

By understanding these four environmental factors and taking steps to ensure that they are all being considered, you can avoid many of the common mistakes that companies make when it comes to business intelligence.

Real-time business intelligence (RTBI)

Real-time BI is a type of artificial intelligence that can analyze and act on data as it comes in, without human intervention. In other words, it can automatically adapt to changing conditions and make decisions on the fly.

Real-time business intelligence (RTBI) is a process that provides business information as it happens, with near to zero latency. It uses data analytics and tools to give organizations insight into their data as it's being produced.

Real-time business intelligence (RTBI) combines data analytics and various data processing tools to enable access to the most relevant, up-to-the-minute data and visualizations. RTBI leverages smart data storage solutions like real-time data warehouses and business intelligence (BI) systems to help organizations make smarter decisions.

Data virtualization, data federation, enterprise information integration, enterprise application integration (EAI) and service-oriented architectures (SOA) are technologies that enable RTBI. Complex event processing tools analyze data streams in real time and either trigger automated actions or alert users to patterns and trends.

To maximize their RTBI tools, organizations need robust infrastructure built to store and process data. They must also have processes in place to help understand the value of their data and have a strategy to collect and analyze the most suitable data types. Strategies may involve building and using a data lake, data warehouse or data mart, or even a combination of these, depending on business needs.

What are the benefits of real-time business intelligence?

RTBI tools come with several benefits that are helpful, out of the box, for different industry verticals. These benefits include the following.

Informed decisions: RTBI can help organizations make better decisions that lead to positive outcomes.

Improved communication: RTBI can help improve communication between teams.

Competitive advantage: RTBI can help businesses gain a competitive advantage.

Real-time alerts: RTBI can provide real-time alerts to avoid delays from batch analysis.

Correlating data: RTBI can correlate business and IT data to pinpoint root causes.

Real-time actions: RTBI allows actions to be propagated back into business processes in real-time.

Instant decision-making. RTBI supports instant decision-making, which is necessary, for example, if a company sells clothing online. The company's website and representatives at the company's call center need to have the same up-to-the-minute data regarding inventory levels. If a customer places an order and a particular size or color is sold out, the customer can be notified and redirected to another, similar item.

Competitive advantage. As consumer expectations grow higher each day, making intelligent decisions based on real-time data is increasingly critical to business relevance. RTBI provides the information organizations need, including forecast analysis and trend analysis, to make tactical decisions that help them take advantage of events as they occur. By using enterprise data to improve customer engagement, productivity and efficiency, businesses can gain competitive advantage and increase revenue. For example, RTBI at Continental Airlines helped transform its industry status from "worst to first" and from "first to favorite." As the Continental Airlines example underscores, organizations cannot ignore the importance of RTBI in overall customer satisfaction.

Rapid detection of critical issues. RTBI isn't just about improving your bottom line. Businesses can also use it as a service intelligence tool to optimize maintenance protocols and prevent potential downtime.

Given the many uses and benefits of RTBI, it's wise to apply a multipronged approach. Some other RTBI application areas include the following:

- application performance monitoring (APM)

- customer relationship management (CRM)
- data security monitoring
- data validation
- demand sensing
- dynamic pricing and yield management
- fraud detection
- operational intelligence (OI) and risk management
- systems monitoring

What Are the Enterprise Applications of Business Intelligence?

Business Intelligence refers to technologies that help organizations gather business data, make an analysis, and present the findings in a form that the relevant stakeholders can easily perceive. The insights from data and visuals help organizations make informed decisions and effective strategies.

Businesses that use business intelligence get to make informed, quick decisions. Given the competitiveness of today's business landscape, businesses must leverage business intelligence applications to stay ahead of their industrial counterparts.

Given the intense competition in today's marketplaces, BI applications remain the most sought-after. Organizations are on a spree to leverage the benefits of BI and stay relevant to the industry and market demands. From finance to retail and gambling, the use of Business Intelligence applications is spread across a broad spectrum of industrial domains, and some of them are listed in the following section.

1. BI for the Finance Industry

The BFSI (Banking, Financial Services, and Insurance) domain endures several challenges, from intense competition to constantly evolving customer demands. Business intelligence techniques can help firms spot emerging trends and patterns, monitor the effectiveness of their products and services, mitigate the risk of unknowns, and build new investment strategies.

Banks, and firms in the financial services sector can automate the end-end loan application and processing process. Using relevant BI apps, customer data can be collected from different data sources, stored, and made accessible to the relevant stakeholders. Doing so prevents silos and enhances transparency in the process. Similarly, financial firms can leverage Business Intelligence applications to mitigate the risks involved in many of their products and services. Based on customer transactions' history, earning capacity, and financial holdings, firms can estimate the risk of granting loans or other debt instruments like credit cards. Business Intelligence applications can monitor the performance and revenue data of different products and services. Based on the data collected, the firms can identify those offerings that are profitable and those that are not. Asset managers can perform sentiment analysis using customer data and build investment strategies.

2. BI for Manufacturing

Using the right BI apps helps manufacturing businesses enhance operational efficiency. Business Intelligence apps help stakeholders gain a complete picture of their supply chain. The analytics derived from such tools help businesses make reliable forecasts and manage inventory. Doing so reduces the lead time and enhances the capability of the firm to meet just-in-time demands.

With data on the workstation capacity and human resource availability, BI tools help manufacturing firms schedule their production process to effectively utilize the available resources. As stakeholders can gain a complete view of the supply chain operations, firms can track costs and work out ways to minimize them.

Also, BI tools help monitor the downtime of employees and the machinery and equipment in a manufacturing firm. The data helps them undertake predictive maintenance and ensure continuous, uninterrupted operation. Based on the data available, firms can monitor the manufacturing cycle of a product and work out ways to reduce the length of the cycle.

3. BI for Online Gambling

Online gambling has been experiencing remarkable growth recently. Casino operators use BI tools to gain insights into which games are doing well and what they can do to reach more players. Online casinos operate worldwide and can use data and analytics to understand where your games are popular. Also, understanding players' interests helps online gambling companies design the best games.

Countries are introducing stringent rules to regulate the online gambling industry. In such cases, the casinos must acquire licenses from legal authorities, which must be renewed periodically. BI tools help prepare regulatory reports at a quicker pace and simplify the compliance needs of the business.

Also, as BI applications track real-time data, the IT department can monitor the data, identify irregularities, and sort them out so the customers can enjoy an uninterrupted gaming experience. Business intelligence techniques helps design bonuses, discounts, and promotions for the players to attract more players.

4. BI for Hospitality

Hospitality businesses deal with a wide range of customers, and retaining them depends on the degree of personalization you offer in the services. Businesses in the hospitality industry depend on business intelligence to make reliable forecasts about occupancy rates. Such forecasts are made while considering several attributes like seasonal changes, trends, and patterns, and so they help businesses maximize their revenue.

Reliable forecasts help hospitality businesses adopt a dynamic pricing model, which has several benefits in terms of revenue optimization. Hospitality businesses also use BI tools to gather booking data through websites, POS systems, CRM software systems, etc.. Using this data, businesses can offer a higher degree of personalization. Similarly, businesses in the hotel industry rely on BI apps to determine which items on their menu are demanded more.

Hotel businesses can use insights from business intelligence techniques and tools to understand when the demand is high. Such forecasts help hotels manage customer demand more effectively. By doing so, hotels can offer discounts for customers who visit the hotel during off-peak hours.

5. BI for Retail

Traditionally, retail businesses relied on spreadsheets and static reports to make decisions. But with BI gaining prominence, businesses can generate dynamic, real time reports and make informed decisions regarding customization, dynamic pricing, floor plan, inventory management, competitor analysis, customer trends, etc.,

Retail firms can use Business Intelligence apps to collect and access data about demographics, sales, purchase history, etc. By processing this information, they can make reliable forecasts about customer demand and manage inventory accordingly. Also, businesses can leverage customer behavior information to personalize their offerings. They can also use the information to build a floor plan so customers can find what they need without hassles.

BI can help retail businesses track their marketing campaign's performance and make necessary changes. They can leverage the features of BI applications to launch and manage stores in multiple locations.

What BI Tools Assist in Applications?

As the previous section shows, BI tools have applications in wide-ranging industries. Each industry listed above can be used to monitor and track different business activities like sales, performance management, reporting, etc. We have covered each of them in detail in the coming section.

1. Sales Intelligence

Sales remains one of the key Business Intelligence applications. Irrespective of the size and scale of the business, the key objective of the sales department in all organizations is to push leads and prospects toward the end of the funnel at a minimum cost.

Businesses can use BI tools to collect data on customer demographics, conversion rate, lifetime value, and other relevant sales metrics. They can organize the data into structured graphs, charts, etc. Businesses can use visual reports to derive trends and patterns in customer behavior and leverage them to make real-time decisions. Such data-driven decisions help businesses generate quality leads, boosting their profits, revenue, conversion rates, etc.

Today's marketplaces are constantly changing and rapidly evolving, and so is customer preference. BI tools help businesses enhance the effectiveness of their sales process. By analyzing trends and data about their leads and prospects, businesses can improve their lead-generation process's efficiency.

For businesses operating in other countries, BI tools can help collect customer data and determine what works for them in each territory. Businesses can use the data to track every sales pipeline activity and identify improvement areas.

Businesses can use BI tools to track historical data on sales trends, purchasing behavior, etc., and use the same to make reliable sales forecast. Doing so helps businesses identify seasonal trends and plan their inventory, which helps them meet the rise in demand during peak season or manage their funds during the off-season when there is no demand.

2. Visualization

Firms generate enormous amounts of data at each point of their business operations which in their original form are unstructured. All your data must be organized to derive insights, and the business's success depend on how best it puts the data to use in real-time. Firms use business intelligence applications to visualize their data in the form of infographics, charts, graphs, etc., which can help stakeholders make business decisions in a quick span of time.

BI tools helps businesses create customized reports based on the KPIs and metrics most important to them. The best thing about visualization is that it presents data in an appealing, easy-to-understand format. This way, even those employees and stakeholders who do not have tech expertise can perceive the data and make informed decisions. Doing so reduces their dependence on data science experts.

Also, studies show that our brains have the power to process images 60,000 times faster than text. So, converting raw data into visuals enhances processing speed and enables quicker decisions. Traditionally, businesses relied on static reports, and new reports must be generated if there is a shock to a variable. Modern

dashboards and visualization tools are interactive, dynamic, and constructed using real-time data.

3. Reporting

As mentioned above, the main purpose of business intelligence applications is to convert original, unstructured data into an organized format. Doing so helps businesses monitor their performance and make changes wherever needed. Businesses rely on several reports, including sales, inventory, customer research, performance, staffing, etc., to devise informed strategies.

Reporting and data analysis are commonly used interchangeably, but their objective and purpose differ. Reporting helps translate data into plain information, while analysis helps turn data into insights. In other words, reporting shows what is happening in the business, while the insights from data analysis show why it is happening. Reporting helps summarize the data, and the insights can be used to enhance the effectiveness of business practices.

4. Performance Management

Business intelligence applications can be used to monitor and track the performance of businesses in real time. Project management teams have targets and deadlines, and businesses can track real-time information to identify gaps and make important changes.

Managers and leaders can use the dashboards and visualizations from BI tools to delegate tasks, set KPIs, and monitor their performance. Such tools can notify employees of upcoming targets and deadlines so they stay in alignment with business objectives. The data can be used to identify the gaps in the process and make changes to enhance the productivity of an organization.

10 Business Intelligence Applications Examples

Business intelligence applications help translate raw, unstructured data into organized information. As the business landscape has become more competitive, businesses are working to stay ahead. So, there is naturally a rise in demand for BI applications, and as a result, several firms have ventured into the space.

To help you understand how you can leverage the BI tools to propel your business goals, we have listed 10 business intelligence applications. If you are exploring examples of business intelligence, this section has all the information you need.

1. Google

Google has been a pioneer in the BI space, for it has a team dedicated to people analytics within its HR department, which is a perfect example of business intelligence. The team focuses on enhancing the processes and productivity based

on daily insights. Their people analytics team extensively researches HR, from leadership to productivity. It doesn't stop here, but it works with the internal department to translate the insights derived from research into action.

Google has leveraged people metrics and data to build a happy, productive workplace. It started with a research study called Project Oxygen in 2008, and the objective was to understand what makes a manager great at Google. The research used data from appraisals, interviews, employee feedback, surveys, etc. The study showed that managers profoundly influenced employee productivity and outcomes.

From the study, Google identified ten behaviors for best managers. The study also revealed that Google employees wanted managers who could promote cross-collaboration and make strong decisions. The study showed that employees wanted managers who were results-oriented and not micromanaging.

2. LendingClub

LendingClub has become the largest online credit marketplace for financing personal, business, medical, and educational purposes. The service provider hired Heap, which helps businesses gain access to customers' journeys and leverage the data to enhance customer conversion, satisfaction, and retention.

The company wanted to run A/B testing to identify the product preferences of a set of customers. With the BI abilities and tools from Heap, LendingClub could run 12 A/B tests a week, significantly higher than the 4 tests it performed a year. The company could test several product combinations, and based on the insights derived from such tests, it tweaked its offerings. Doing so helped the business reach more customers and boost its revenue.

3. Grow

Grow is a business intelligence firm, and it partnered with Heap to figure out what businesses expected during the free trial of its visualization tools.

The study revealed that customers who attempt to create a visualization within the first five minutes of them signing up for the Grow tool are more likely to become their regular, paying customers. This finding was a new revelation to Grow firm which had thought that customers prioritized quality and accuracy more than speed of setup.

Grow initially had a long setup process that took more than 90 minutes, and the users had to integrate their customer, marketing, sales, etc., data into the Grow platform to be able to use the tool. Based on the insight from the Heap business intelligence application, they worked on streamlining the setup process and reducing the time it takes.

4. Stitch Fix

Stitch Fix offers a virtual stylist to its customers, who are expected to complete a questionnaire with 90 data points, including size, height, style, etc. Based on the responses, the company styles the clothes, shoes, and other accessories and ships them to the customers. They can choose and buy what they like and return what they are uncomfortable with.

Initially, the stylist has access to limited data. But with repeated orders and continuous feedback, the business intelligence tools help analyze the data and make more meaningful recommendations to the customer. It uses eigenvector decomposition, a concept from quantum mechanics, to decode the overlapping notes and layers in a customer's style. Doing this requires complex skills and abilities, which would be impossible without reliable business intelligence tools.

5. Target

Target is a popular retail chain housing apparel, accessories, beauty, food and beverages, toys, electronics, and home accessories. Because the store sells everything from home decor to milk and food, it had a hard time convincing customers that the only store they may have to visit to meet their basic needs is Target. It used the principles of habit formation to target customers and dish out personalized marketing offers and managers.

As a part of its marketing strategy, the Target store followed a pregnancy-prediction model. It would assess every female customer based on data points like shopping history and make a pregnancy prediction score. Based on the score, it would send coupons and personalized offers about diapers and cribs.

Back in 2012, the pregnancy-prediction model had estimated high scores for a young girl who was doing high school. When the store started receiving complaints, it apologized and turned out to leverage business intelligence abilities to make more accurate, personalized recommendations without surveilling the subtle nuances of the customers.

6. Payless Shoes

Payless Shoes evolved into one of the USA's largest footwear retailers. The company always worked on enhancing customer experience, and when it started going online, it initially assumed that customers expected a one-click buying experience. When businesses started selling online, all of them were driven by the notion that customers enjoyed shopping from websites with fewer clicks than otherwise.

But after partnering with Heap, the leading Business Intelligence application platform, data from customers of Payless Shoes revealed they preferred a two-click process.

7. Twitter

When Twitter was launched in 2006, it restricted tweets to 140 characters. In 2017, the platform started allowing 280 characters. But analysts and users thought it might backfire because allowing 280 characters may make the posts wordy. They thought users might discard the platform as there may be a drop in meaningful anecdotes and short jokes.

But against the notion, users enjoyed tweeting. As they could add longer tweets than before, they did not have to spend their time adjusting and rephrasing the content to 140 characters anymore. Twitter had brought this change based on the insights from their Business Intelligence tools, whereas users did not have access to such tools. Such tools helped businesses make meaningful decisions.

8. Facebook

Facebook relies on business intelligence to improve the features and functionalities of the platform. Academia has also started showing interest in analyzing the platform's impact on people's lives.

One such study conducted by Stanford University suggested that Facebook usage has a detrimental effect on the mental health of its users. The primary cause was that people tend to compare their lives with that of others and develop a feeling of scarcity. They fail to understand that people on Facebook decide how they wish to be perceived by the community and carefully curate their best version.

People tend to compare the number of likes they have received and someone else has received and form opinions. Based on this observation, the company allowed users to keep their like counts private so only the profile owner could see how many likes their posts received.

9. TikTok

TikTok remains a popular communication and social media platform. The app offers users recommendations based on their preference data to keep them hooked on to the app. You only need to download the app from AppStore or Google Play Store and start watching videos.

When you react to the videos with a hard press or heart symbol, the app's algorithm reads user preferences and makes recommendations. As it recommends videos the users are interested in, they tend to spend more time using the app. Doing so is very important in this era where social media apps strive to increase users' time on their platforms.

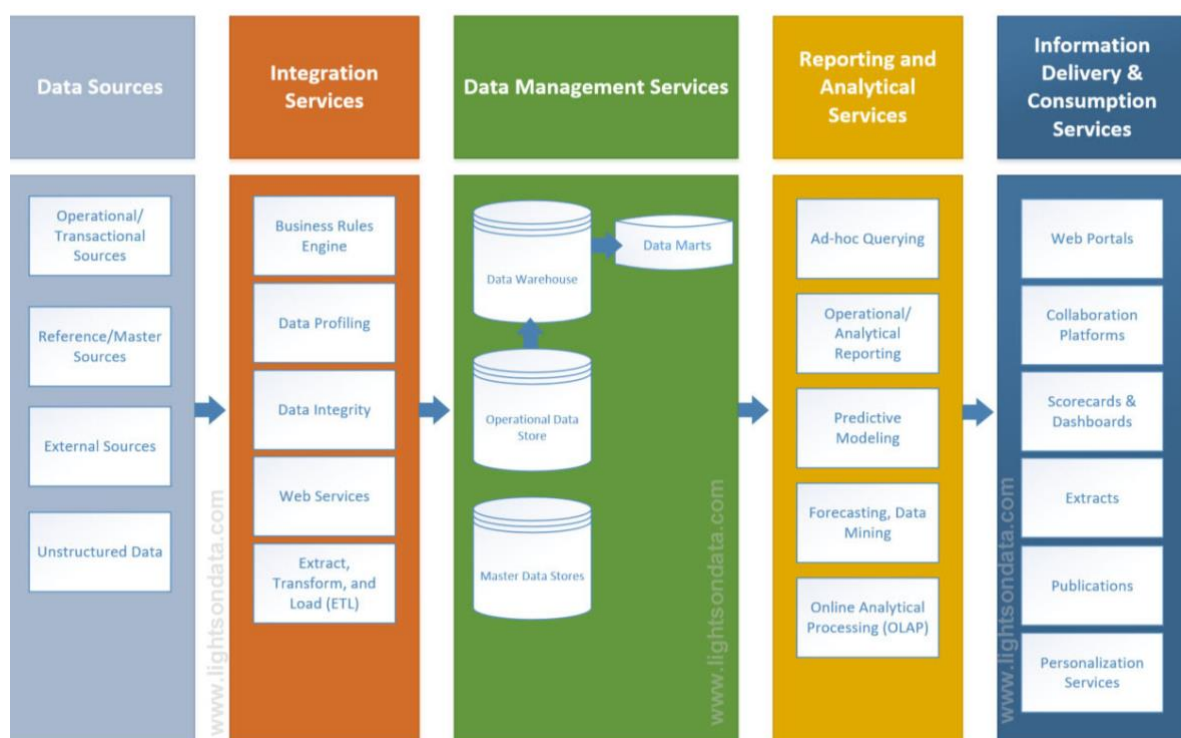
10. Uber

Uber, the robust ride-hailing app, launched its food delivery under the aegis of UberEATS. The company leveraged the power of business intelligence applications to generate data and actionable insights that help propel business objectives.

UberEATS launched Restaurant Manager, a business intelligence app to help restaurant partners gain access to important metrics and KPIs like net order value, revenue, items sold on the app, customer feedback, etc. Based on the data, restaurants can gain insights into what items are most demanded by customers. The partners can also make reliable forecasts, ensure timely delivery, and enhance customer satisfaction.

BUSINESS INTELLIGENCE LANDSCAPE

Data is only meaningful if it serves a purpose. This purpose is to have it transformed into meaningful insight, hindsight, and foresight for an organization to take advantage of. Similarly to how raw materials go through a complex conveyor belt system in a factory before the end product is created, so does the data before it gets transformed into information. I refer to this complex conveyor belt system for data processing as the **business intelligence landscape**, or the BI landscape for short. Understanding this landscape is the first step in putting together a BI program and determining where initial resources should be invested in. The details of each landscape tend to be unique to the industry, business needs, and the organization which it's part of, but most of the time, all business intelligence landscapes have the same format at a macro level. It's usually comprised of 5 pillars and 5 foundation blocks:



Data sources

This pillar outlines the different sources from which your data enters your ecosystem. Typically these are:

- Operational/ transactional sources – Ex: online purchase/donation forms

- Reference/ master sources – Ex: HR information system, reference master data (such as fund types)
- External sources – Ex: Neighborhood wealth ratings database, National Change of Address services
- Unstructured data – Ex: documentation, emails, phone conversation recordings

You need to be aware of all these sources and how data is captured from each one in order to evaluate the data quality level and ensure the proper transformation and business rules are assigned before it can be stored within the systems you manage.

Data integration

Also known as the “data gate keeper pillar” as it evaluates, standardizes, updates, and transforms the data through:

- Business rules & procedures – create and optimize them and ensure the data follows them
- Data profiling – a step often missed, but of great value in understanding the data better and gauging its quality level
- Data integrity – ensuring data is recorded or captured as intended
- Data manipulation – the actual process of importing it and integrating the data into your systems

Data management

This refers to your typical systems housing the data once under your scope. It consists in the maintenance of:

- Master data store(s)
- Operational data store(s)
- Data warehouse and/or data marts
- Potentially even shadow DBs consisting of Excel files

Reports and Analytics

This core BI pillar is where the data gets outputted and/or transformed into information through:

- Ad-hoc queries
- Operational/ analytical reporting
- Predictive modeling
- Forecasting & data mining

- Online analytical processing
- & other

Information dissemination

The pillar through which the created information reaches your intended audience. This is usually done through:

- Web portals – Ex: an intranet through which user access is maintained
- Collaboration platforms – Ex: a SharePoint site
- Dashboards/ scorecards – Ex: PowerBI or Tableau
- Extracts – Ex: Excel data sheets
- Publications – Ex: Executive summaries, board reports, organization or department newsletters
- Personalization services – Ex: a report automatically delivered through an email, or a dashboard dynamically tailored to its user

A business intelligence (BI) platform is a system that helps businesses gather, analyze, and visualize their data to improve decision-making.

BI platforms can help businesses:

- Make better decisions

BI platforms help businesses use data to make better decisions by turning data into a competitive advantage.

- Visualize data

BI platforms offer tools like dashboards, charts, and graphs to help users visualize large amounts of data.

- Customize

BI platforms can be customized to meet a business's needs, allowing users to choose the KPIs and visualizations that are most relevant.

- Integrate

BI platforms can integrate with other business tools, such as customer relationship management (CRM) and enterprise resource planning (ERP) software.

Some BI platforms include:

Tableau, Microsoft Power BI, Qlik Sense, Yellowfin BI, Sisense, Zoho Analytics, Domo, Datapine, Looker, and DataBox.

What are the four types of business analytics?

The four subsets of data analytics are descriptive, diagnostic, prescriptive, and predictive. Businesses across all types of industries utilize these specialty areas in analytics to increase overall performance at all levels of operations. These four types of analytics all work together and can be used together to help improve business performance.

Descriptive Analytics

Descriptive analytics can show “what happened” and is the foundation of data insights. According to Investopedia, it is the interpretation of historical data to better understand changes that have occurred in a business. This type of analytics can be used to gain an overall picture of how a business is performing and is often used alongside predictive and prescriptive analytics. Common insights include year-over-year comparisons, the number of users, and revenue per subscriber.

Diagnostic Analytics

Diagnostic analytics addresses “why things happened.” Common diagnostic analytic techniques/insights include drill-down, data discovery, data mining, and correlations.¹² Companies use this data to identify patterns of behavior and make deep connections within the data they have collected. In order to be effective, diagnostic data must be detailed and accurate.

Predictive Analytics

Businesses use predictive analytics to “see the future” and predict “what is likely to happen.” Existing data, modeling techniques, and statistical modeling are leveraged to generate predictions about performance and future outcomes. Predictive models are especially useful for marketing and insurance companies which need to make decisions based on what could be coming up.

Common processes in predictive analytics include decision trees, neural networks, and regression models.¹³ Compared to descriptive and diagnostic

analytics, which is fairly common in most businesses, predictive analytics is more intensive and many companies are not leveraging this type of analytics yet.

Prescriptive Analytics

Prescriptive analytics, analytics driven by AI (Artificial Intelligence) systems, helps companies make decisions and determine “what they should do next.” This is the most in-demand type of analytics today, however, it is talent and resource-expensive: Few companies have the skilled employees and resources to conduct it.

This type of analytics is on the leading edge of the analytical landscape and requires sufficient investment and commitment across the entire organization that wishes to perform it. Big data players like Apple, Netflix, and Facebook are currently conducting prescriptive analytics successfully. AI itself falls within the category of prescriptive analytics. It requires tremendous data and continuously updated data to help it learn, refine its decisions, and then communicate and act on these decisions in a business setting.

Business Analytics Applications

Businesses today are faced with two very stark realities—the world is hyper-competitive, and data drive it. Companies that have the best information make the fewest mistakes, which in turn helps them to stay ahead of the pack.

Today's digital society, through the explosion of Big Data and the Internet of Things (IoT), has produced a ton of information. The challenge is to make any sense of all this data. With all of that information, who can sort out what's useful and what's not? That's why business analytics is essential for today's industries, and business analysts are in high demand. Today, we're taking a look at popular business analytics applications, and some of the often-used cases.

Business Analytics Defined

Before launching into the meat of the matter, let's take a moment to review. What's the definition of business analytics? Business analytics involves the collating, sorting, processing, and studying of business-related data using statistical models and iterative methodologies. The ultimate goal is to glean practical and actionable business insights to solve an organization's problems—boosting efficiency, productivity, and revenue.

Note that there's a difference between business analytics and business intelligence (BI), though they are related. Business intelligence falls within the discipline of business analytics, the process of gathering the needed data from all sources, and preparing it for use by business analysts. In short, BI tells you what's going on, and business analytics tells you why it's happening and when it will occur again. So, a business analyst identifies a company's weak areas, collects and sifts through data, creates a plan based on those findings, and helps to implement it.

Applications of Business Analytics in Various Industries

Although business analytics is being leveraged in most commercial sectors and industries, the following applications are the most common.

1. Banking

Credit and debit cards are an everyday part of consumer spending, and they are an ideal way of gathering information about a purchaser's spending habits, financial situation, behavior trends, demographics, and lifestyle preferences.

2. Customer Relationship Management (CRM)

Excellent customer relations is critical for any company that wants to retain customer loyalty to stay in business for the long haul. CRM systems analyze important performance indicators such as demographics, buying patterns, socio-economic information, and lifestyle.

3. Finance

The financial world is a volatile place, and business analytics helps to extract insights that help organizations maneuver their way through tricky terrain. Corporations turn to business analysts to optimize budgeting, banking, financial planning, forecasting, and portfolio management.

4. Human Resources

Although HR is often the punchline of many office jokes, its value in keeping a company successful is not to be underestimated. Great businesses are composed of a great staff, and it's HR's job to not only find the ideal candidates but keep them on board. Business analysts help the process by pouring through data that characterizes high performing candidates, such as educational background, attrition rate, the average length of employment, etc. By working with this information, business analysts help HR by forecasting the best fits between the company and candidates.

5. Manufacturing

Business analysts work with data to help stakeholders understand the things that affect operations and the bottom line. Identifying things like equipment downtime, inventory levels, and maintenance costs help companies streamline inventory management, risks, and supply-chain management to create maximum efficiency.

6. Marketing

Which advertising campaigns are the most effective? How much social media penetration should a business attempt? What sort of things do viewers like/dislike in commercials? Business analysts help answer these questions and so many more, by measuring marketing and advertising metrics, identifying consumer behavior and the target audience, and analyzing market trends.

Business analytics plays a valuable role in many different industries. Its also noticed that some of the applications merge into each other, but that's hardly surprising. By leveraging business analytics, multiple departments and teams can coordinate their efforts based on the information gathered and processed. It's up to the business analyst to identify roadblocks and areas that need improvement, helping different departments to work together to achieve a common goal.

Business Analytics Applications

1. Customer Segmentation

Customer segmentation is a vital business analytics application that helps companies group their customers based on shared characteristics such as demographics, buying behavior, and preferences. By analyzing customer data, businesses can tailor their marketing strategies, product offerings, and customer service to target specific segments effectively, increasing customer satisfaction and overall profitability.

2. Predictive Analytics

Predictive analytics leverages historical and real-time data to forecast future trends and events. This application is used extensively in industries like finance, healthcare, and e-commerce for tasks such as predicting stock prices, patient outcomes, and product demand. It enables proactive decision-making, risk mitigation, and optimization of business operations.

3. Supply Chain Optimization

Businesses utilize analytics to optimize their supply chains by analyzing data related to inventory levels, supplier performance, transportation logistics, and demand forecasting. By identifying inefficiencies and bottlenecks in the supply chain, companies can reduce costs, improve product availability, and enhance overall operational efficiency.

4. Fraud Detection

Fraud detection analytics employs advanced algorithms and machine learning models to identify and prevent fraudulent activities, such as credit card fraud, insurance fraud, and cyberattacks. By analyzing transactional data patterns and anomalies, organizations can minimize financial losses and maintain the trust of their customers.

5. Market Basket Analysis

Market basket analysis involves examining customer purchase history to discover patterns in product co-purchases. Retailers use this application to optimize product placement, cross-selling, and promotional strategies. By understanding which products are frequently bought together, businesses can increase sales and enhance the customer shopping experience.

6. Churn Analysis

Churn analysis focuses on identifying and reducing customer churn, which is the rate at which customers stop using a company's products or services. By analyzing customer behavior and feedback, businesses can implement retention strategies to retain valuable customers and reduce revenue loss.

7. A/B Testing

A/B testing is a fundamental analytics application for optimizing digital marketing campaigns and website performance. It involves conducting controlled experiments by randomly assigning users to different versions of a webpage or marketing content. By comparing the performance of these versions, companies can make data-driven decisions to improve conversion rates and user engagement.

8. Employee Performance Analytics

Employee performance analytics helps organizations evaluate the productivity and engagement of their workforce. By analyzing data on key performance indicators (KPIs), attendance, and employee feedback, companies can make informed decisions about talent management, training, and workforce optimization.

9. Quality Control and Process Improvement

In manufacturing and production industries, analytics is employed to monitor product quality, detect defects, and optimize production processes. By analyzing data from sensors and production lines, businesses can reduce defects, improve efficiency, and minimize waste.

10. Sentiment Analysis

Sentiment analysis, also known as opinion mining, uses natural language processing and machine learning techniques to assess public sentiment and

opinions from sources like social media, customer reviews, and surveys. Companies can gain insights into how their brand is perceived and use this information to shape marketing strategies and product development.

Usage of Business Analytics

Business analytics helps organizations run more efficiently and profitably. Here are six cases where business analytics proves its worth in the commercial sector.

1. Churn Prevention

Churn is the customer attrition rate, a percentage of subscribers, or customers who stop doing business with a company. Successful companies must keep the churn rate low and replace any customer losses that inevitably occur. Furthermore, it's more expensive to acquire new customers than it is to retain existing ones. By using predictive analysis, a business analyst helps identify customer dissatisfaction and the most likely risks or departure.

2. E-Commerce Personalization

Online businesses, like Amazon, collect, process, and analyze customer data to personalize their customers' shopping experiences. By customizing the experience, vendors can make recommendations and increase the likelihood of further sales.

3. Predictive Maintenance

Companies must face the inevitability of equipment maintenance, both scheduled and unplanned. Business analysts work with data to create metrics about maintenance lifecycles to predict future maintenance needs and avoid costly unplanned downtime.

4. Insurance Fraud Detection

Insurance fraud is costly to companies and their customers alike. This is especially true in the medical insurance industry, where fraud costs organizations in the US approximately \$68 billion a year. Business analysts use big data to process billions of claims and billing records, enabling investigators to identify and mitigate any fraudulent activity.

5. Automated Candidate Placement

As mentioned earlier, hiring new staff comes with its share of risks and uncertainty. Business analysts leverage data-driven recruitment platforms to get a better picture of any given candidate—improving the likelihood of a successful job match much faster. In some cases, the information can even help anticipate job needs before a position is posted.

The Emerging Need for Business Intelligence

This vital **role and need of Business Intelligence** (BI) in modern organizations, emphasizing how it empowers data-driven decision-making, fosters efficiency, and ensures competitiveness and compliance.

1. Competitive Advantage:

In the fiercely competitive contemporary business landscape, maintaining a competitive edge is a perpetual struggle. **Business Intelligence** (BI) emerges as the game-changer, bestowing organizations with the tools to make data-driven decisions.

Through meticulous **analysis of market** trends, insightful understanding of customer behaviour, and precise assessment of competitor performance, businesses can refine their strategies, strategically reposition themselves, and effectively outperform their rivals.

In a world where information is power, BI is the beacon guiding businesses towards the path of strategic excellence and sustainable success.

2. Improved Decision-Making:

In the realm of successful organizations, sound decision-making stands as the pivotal foundation. The dynamic landscape of business demands swift and precise judgments. This is where **Business Intelligence** (BI) emerges as the guiding light.

It equips decision-makers with the indispensable arsenal of data-driven insights, enabling them to meticulously evaluate risks, uncover latent opportunities, and judiciously allocate resources.

With BI at their disposal, the ambiguity of guesswork dissipates, paving the way for leaders to navigate uncharted waters with confidence, ultimately ensuring that their choices are founded on a bedrock of well-informed wisdom.

3. Enhanced Operational Efficiency:

Business Intelligence (BI) serves as the ultimate efficiency booster for businesses. It meticulously scrutinizes production data, supply chain logistics, and employee performance. This analytical prowess uncovers hidden inefficiencies and bottlenecks, offering organizations a roadmap to streamline their operations.

By **addressing these pain points**, businesses not only cut costs but also supercharge productivity. In the quest for optimization, BI becomes the trusted ally, helping businesses navigate the intricate labyrinth of data and emerge leaner, more agile, and incredibly efficient.

4. **Customer Insights:**

Understanding and fulfilling customer needs is the cornerstone of business success. In the dynamic market landscape, **Business Intelligence** (BI) tools are indispensable. They dissect vast troves of customer data, extracting invaluable insights.

This treasure trove of information empowers businesses to customize their offerings, aligning perfectly with the nuanced preferences of their target audience. By doing so, companies not only enhance customer satisfaction but also boost brand loyalty and profitability, ultimately achieving sustainable growth in the fiercely competitive business world.

5. **Risk Management:**

In a constantly changing world full of uncertainties, the **management of risks** is crucial. **Business Intelligence** (BI) serves as a beacon in this sea of uncertainty, spotlighting potential risks and offering early warnings.

Armed with these insights, organizations can proactively strategize and implement measures to minimize or even prevent the impact of these risks, safeguarding their stability and ensuring they stay ahead in an unpredictable business landscape.

6. **Compliance and Governance:**

Countless industries face stringent regulatory demands. **Business Intelligence tools play a pivotal role** in upholding compliance by meticulously monitoring and documenting all facets of their operations. This not only guarantees adherence to legal and ethical guidelines but also minimizes the risk of regulatory breaches.

Businesses can rest assured that their practices are in line with industry standards, safeguarding their reputation and avoiding potentially costly

penalties. **Business Intelligence** acts as a guardian, ensuring organizations stay on the right side of the law while simultaneously promoting an ethical and responsible business environment.

The Roles of Business Intelligence

The Roles of **Business Intelligence** encompass **data** analysis, development, and advanced analytics by data scientists, serving business users across organizations, enabling data-driven decision-making and performance optimization. Let's throw more light on these roles in the following text;

• Data Analysts:

Data analysts are the linchpin of Business Intelligence (BI). Their pivotal role involves not only collecting and processing data but also interpreting it to uncover valuable insights. They act as data alchemists, transforming raw information into actionable knowledge. By delving into the depths of data, they reveal hidden patterns, trends, and correlations. Equipped with this newfound wisdom, they serve as the bridge between data and decision-makers, presenting clear and concise reports. In this symbiotic relationship, data analysts empower organizations to make informed choices, **driving growth and efficiency** in an increasingly data-driven world.

• BI Developers:

Business Intelligence (BI) developers are the architects behind the data-driven revolution. They build the foundation for data collection and analysis, crafting intricate databases that house valuable information. These experts go beyond mere **data storage; they fashion dynamic dashboards, transforming raw data into insightful visuals that guide decision-makers.** BI developers play a pivotal role in ensuring data accessibility, making sure that the right information is available to those who need it, and fostering a culture of informed decision-making within organizations.

• Data Scientists:

Data scientists elevate Business Intelligence to new heights by employing cutting-edge analytics and harnessing machine learning capabilities. Their expertise enables the discovery of concealed patterns within data and empowers the generation of predictive insights rooted in historical data. This specialized skill set enables organizations to gain a competitive advantage, make proactive decisions, and anticipate future trends, ultimately steering them towards informed and strategic actions.

- ***Business Users:***

Business Intelligence (BI) is a game-changer for organizations. Employees, regardless of their position, experience the transformative impact of BI. Through user-friendly BI tools, they gain seamless access to vital reports and interactive dashboards, granting them the power to base their daily decisions on data-driven insights. This not only enhances efficiency but also fosters a culture of informed decision-making, ultimately steering the organization towards greater success in the ever-evolving business landscape.

Importance of Business Intelligence in Financial Institutions

Financial institutions, in particular, reap substantial **benefits from Business Intelligence**. They rely on accurate and up-to-date information to make investment decisions, **manage risk**, and comply with strict financial regulations. BI enables financial institutions to:

- ***Risk Management:***

Financial **institutions must prioritize effective risk management** to ensure their stability and success. By thoroughly analyzing market data, credit scores, and customer behavior, organizations can proactively pinpoint potential threats and take preventative measures.

This ensures stability, protects investments, and maintains trust among clients, ultimately safeguarding the institution's long-term success.

- ***Fraud Detection:***

Fraud Detection: Safeguard your organization by actively monitoring transaction data for any irregularities, swiftly identifying suspicious activities, and taking proactive measures to prevent fraud. This crucial component of **Business Intelligence** ensures financial security and maintains the trust of your stakeholders.

- ***Customer Relationship Management:***

Enhance customer relationships through comprehensive Customer Relationship Management (CRM). This involves gaining profound insights into customers' financial requirements and preferences. By tailoring products and services to meet these needs, businesses can foster loyalty, trust, and long-term satisfaction, ultimately driving sustained growth and success.

- ***Compliance:***

Compliance is paramount, ensuring strict adherence to complex **financial regulations and meticulous reporting** standards to maintain legal and ethical integrity.

- ***Investment Analysis:***

Investment Analysis is a pivotal aspect of financial decision-making. It involves a meticulous examination of market trends and economic indicators, enabling investors to make well-informed choices that optimize returns and minimize risks in their portfolios.

Enterprise Business Intelligence

Ask any leader today if they want their organization to be more data-driven, and the answer is likely to be yes. In fact, this demand for more insights at every level of an organization is one of the top trends in analytics, especially as we face continued economic uncertainty.

Making informed business decisions is no longer a luxury – it's a requirement for every business, in every industry. The world of business analytics can provide meaningful, valuable insights that can help organizations make better decisions and create more value. The challenge, however, lies in unearthing these insights from all various data sources, and presenting them in an understandable, meaningful way - often through data storytelling - that enables the business to take effective action quicker than ever before.

This is where enterprise business intelligence (BI) comes into play. From providing access to interconnected data sources to helping visualize results for easy consumption, these comprehensive platforms offer value at every turn for businesses of all sizes and industries alike. Read on as we discuss what makes up enterprise BI.

Enterprise business intelligence (BI) is a type of software solution that uses data analysis and other tools to provide users with insights into their businesses. It helps organizations make better decisions, improve operational efficiency, reduce costs, provide more accurate sales forecasts, and gain competitive advantage by infusing data into decisions across the board. BI solutions can be used to analyze customer data, market trends, sales performance, financial information, production data, and

more, turning this data into meaningful insights that can inform a company's strategies.

What can the best Enterprise Business Intelligence platforms do?

Enterprise BI platforms have been around for years. While the first generation of tools focused on perfect reporting (think Cognos and Business Objects), the second generation of tools focused on visualizing this data. Today, however, the most valuable enterprise business intelligence tools go beyond visualizing data, empowering everyone, not just data professionals, to serve their own insights through familiar paradigms like search and AI.

Here are some of the top features you should look for in a modern enterprise business intelligence platform:

Search-driven analytics

We all use search daily in our personal lives to ask and answer questions. The best enterprise BI tools allow users to do the same with data, so users can search using natural language queries. These queries are then converted into SQL queries and run against the underlying data sources. This allows users to quickly and easily explore their data and get insights without needing to know complex SQL queries. BI and data teams are then freed from endlessly updating dashboards and reports, and able to focus on more strategic data initiatives such as analytics engineering and data modelling

AI-powered insights

Simply getting an answer to the questions you know isn't enough for BI tools today. A new must in enterprise business intelligence is using artificial intelligence and machine learning algorithms to automatically suggest insights and recommendations based on user queries, usage patterns, and other behaviors from other similar users. This can help users identify trends, anomalies, and correlations in their data that they may have missed otherwise. With tools like SpotIQ, you can instantly get suggestions on additional searches or further explorations as you go with the power of AI. As people search to analyze their data in ThoughtSpot, SpotIQ learns what's important and tunes its algorithms to provide more relevant insights.

Provides self-service

In the decade of data, data is everyone's job. Self-service BI capabilities allow all kinds of users, not just those who know SQL or programming languages, to ask and answer

their own questions without relying on IT for assistance. By saving data analysts time and IT resources, analysts can work on more value-add projects. This empowers users across the enterprise to access and analyze data, leading to improved collaboration and decision-making in a much faster way. The most powerful of these tools, such as ThoughtSpot Sage, provides this self-service in natural language, so users can interact with data as easily as they chat with a colleague.

Interactive visualization

Once you've found insights, having a BI platform that can present these through interactive visualizations in a Liveboard instead of a static dashboard is critical to taking insights and turning them into action everyone can understand. This enables users to easily create charts like bar graphs and histograms, charts, and tables to represent their data and gain a better understanding of trends and patterns. By making them interactive, you can explore or drill into data, right from the visualization, until you or your users find the answer they need.

Data governance and security

There's a reason data security and governance remains top of mind for data leaders. An absolute must for enterprise business BI platforms is providing robust data governance and security features, including role-based access control, data lineage, and audit trails. This ensures that sensitive data is protected and only accessible to authorized users.

Data integration

Tools that cannot work natively with your cloud data platform are going to hold your BI strategy back. Pick a BI platform that integrates with a variety of data sources, including databases, cloud data warehouses, data lakes, and any other kind of data your business relies on. This allows users to easily access and analyze data from multiple sources without having to switch between different platforms. The best platforms like ThoughtSpot offer this as a native connection, so no data is cached or copied, meaning insights are always as fresh and accurate as the underlying data.

Collaboration

Business intelligence is not a solo sport. You will want an enterprise BI platform that provides collaboration features that enable users to share insights, reports, and other analytical content with others. Users can also collaborate in real-time and leave comments and feedback on shared content, including being able to collaborate outside the BI platform in tools like Slack and Teams.

Top 5 benefits of using enterprise business intelligence

There are numerous benefits to implementing enterprise business intelligence within an organization. While companies have sought these benefits for years, legacy BI tools have made them more a vision than a reality.

However, with the rise of cloud ecosystems delivering the modern data experience with an enterprise BI platform like ThoughtSpot is finally possible. Here are six key benefits of doing so:

1. Improved decision-making

Modern enterprise BI provides decision-makers with real-time, accurate, and relevant information to make informed decisions, when and where those decisions are made. This can lead to faster decision-making and a more agile organization overall. This information is presented in an easily understandable format, such as charts, graphs, and tables, which help executives quickly identify trends, patterns, and anomalies that can lead to new or different courses of action. This enables them to make informed decisions that are backed by data and insights, rather than relying on intuition or guesswork.

2. Better insights into business performance

With the right enterprise BI solution, leaders can gain a comprehensive view of business operations and performance. By analyzing data from various sources, decision-makers can identify areas where they are excelling, as well as areas where they need to improve. This can lead to increased efficiency, reduced costs, and improved profitability. It's critical to have a true, single source of truth in your data to avoid conversations of whose view into business performance is better, so look for BI tools that let you view all your business data in its entirety, instead of requiring you to cube, aggregate, summarize, copy, or cache it.

3. Increased efficiency and productivity

In business, time is money. Enterprise BI that's built on self-service can help automate routine tasks and streamline business processes, giving data teams time back to work on more important, strategic projects while empowering business users to immediately ask and answer their own questions. This helps everyone get more time to focus on their areas of expertise without sacrificing data-driven decision making.

4. Improved customer satisfaction

At the end of the day, the best way to delight customers is to understand them. By analyzing customer data, enterprise BI can help organizations identify trends and patterns in customer behavior. This can lead to improved customer service, targeted marketing campaigns, and increased customer loyalty.

5. Competitive advantage

With shorter and shorter cycles of creative destruction, every ounce of competitive advantage matters. With the right enterprise BI, organizations can make data-driven decisions faster and more accurately than their competitors. By gaining insights into their business operations and industry trends, companies can identify new opportunities, make better strategic decisions, and optimize inventory and processes.

EBI as a Revolution

Enterprise business intelligence is the key to business performance, allowing decision-makers to immediately access information and act on it in a timely manner. With the right platform, decision makers can access data in real-time, find insights quickly and accurately, as well as have access to data that generates deeper value. Furthermore, using enterprise business intelligence also has various benefits such as increased efficiency, improved accuracy of decision making, cost savings and improved customer service.

The right solution, however, is essential in capturing these benefits and avoiding the pitfalls that have kept the BI industry from delivering on its promise of data democratization. Tools that can connect directly to the modern data stack, empower business users not just data professionals, and let you leverage the best in data modeling are critical. ThoughtSpot is the leader in AI Analytics, bringing all of these capabilities to your data and taking your BI efforts to the next level. If you're looking for an enterprise business intelligence platform that's easy to use, try out ThoughtSpot's free trial today. Start revolutionizing the way you get insights from your data.

Emerging Trends

What the Future of Business Intelligence Holds

Like every other technology, the latest BI trends will continue to evolve. From the early days when BI was confined to spreadsheets bursting with numbers, to today, where technology allows for insightful visualization and immediate action. Today, BI provides organizations with new and novel ways to improve productivity, increase profits, and understand customers. As technology advances with the blink of an eye, the future of BI looks extremely promising. However, to realize the real value from BI, it is important to be aware of upcoming trends, understand if and how they can be incorporated, and have a roadmap in place to embrace technology across the business.

1. Data Governance

As regulations across data privacy and security become more stringent, data quality management has become indispensable for business excellence. The coming years will see organizations driving sustained efforts towards classifying data, knowing where it is coming from, who has access to it, how people are using it, and how long they can keep it.

In the realm of BI, data governance will become a top priority for organizations big and small. This will also be driven by the number and complexity of data sources and data types needed to support analytics initiatives that are increasing exponentially. A sound data governance strategy will not only help improve ROI from BI investments; it will also enable a healthy balance between data consistency and transparency. This will, in turn, set the foundation of accurate, ethical, and evidence-based decision-making.

Data governance will enable organizations to clearly understand the information needs of the enterprise. This will continually improve the quality of data and information while also ensuring privacy, confidentiality and will prevent unauthorized use. It will allow organizations to use the right data to guide informed BI decision-making and ultimately improve business outcomes. It will also ensure data is acquired securely from approved sources, processed for the intended purpose, shared with authorized personnel, and discarded as per a pre-defined timeline.

2. Self-service BI

As business users get increasingly tech-savvy, they expect anytime, anywhere access to the data they need to do their jobs well. They want to be able to solve

issues on their own – without having to depend on an analytics team. With centralized data from across the organization, users want to use the tool of their choice to drive value.

Although self-service BI is already popular, it will become a global norm in the coming years, enabling users to get the insight they need to tackle challenging business problems in a timely fashion. It will also help optimize the decision-making capability across the organizations while powering users with a single version of the truth – in whatever they do.

Since business users know what they want, having a self-service data model will enable them to become self-sufficient. The coming years will witness users make better use of data and pave the way for an organization-wide data culture. Self-service BI will break the dependency on IT/data teams to get access to the right data, and users will be able to meet their analytical requests with ease and make critical decisions at a much faster pace.

3. Prescriptive Analytics

Although predictive analytics has been an integral aspect of BI that enables organizations to extract information from existing data sets in order to forecast future probabilities, the coming years will witness prescriptive analytics becoming a worldwide feature.

Prescriptive analytics will examine data or content to determine what steps need to be taken to achieve an intended goal. By gauging the impact of future decisions, it will enable organizations to adjust the decisions before they are made – thereby improving decision-making accuracy.

For the next couple of years, organizations will use prescriptive analytics to factor in information about possible situations, available resources, past performance, and so on and get suggestions on the best course of action or strategy. They will understand worst-case scenarios better and make decisions based on highly analyzed facts – rather than jump to conclusions based on instinct.

4. BI through Natural Language Processing (NLP)

As the relationship between humans and machines becomes more and more profound, the growth of NLP is unprecedented. Although smart voice assistants have long been responding to users through NLP, enterprises will also improve business results to pursue a more significant competitive advantage in the coming years. In a few years, NLP will be used not just in the customer service department, but also in other business areas. For example, NLP will allow not-so-savvy employees to make sense of complicated systems and digital tools. It will also

make BI-based data more accessible and enable non-technical users to connect to data – quickly and easily.

Using NLP, businesses will analyze customer sentiments, abstract information from a piece of text, or determine how positive (or negative) social media buzz around them is. The technology will also evolve into AI-powered chatbots, providing quick and accurate answers to users' BI inquiries.

5. Business Intelligence-as-a-Service

As the XaaS model permeates into virtually every area of business, the realm of BI and data analytics will be no exception. Companies with humongous amounts of data, that have difficulty accessing the data or driving insights from it, will look out for BI-as-a-Service options. This model will deliver all the benefits of a full-fledged, end-to-end BI solution, but with the ease and simplicity of a cloud deployment.

BI-as-a-Service will enable organizations to get a BI solution up and running in no time while freeing their IT staff from carrying out complex analysis tasks. The model will allow organizations to get ready access to expert BI consultants and data architects who understand data and help manage and govern it to drive better business results – at substantially low costs. Using BI-as-a-Service, organizations can have experts seamlessly extract data from multiple sources, organize and analyze the growing volume of data, and present insights to users using intuitive dashboards and reports. The end-to-end service will not only enable them to overcome bottlenecks around their small number of data scientists, but it will also allow everyday users to create reports and dashboards on their own.

6. Collaborative and Integrative BI

The coming years will witness collaborative and integrative BI become increasingly mainstream. Unlike today when stand-alone BI tools must be purchased and implemented separately, most enterprise systems will have built-in BI capabilities, so users can unearth insights and make decisions – without ever leaving the platform.

The next-generation BI systems will be geared toward larger sets of users and more connected to bigger enterprise systems. They will continuously pull data from the required sources, consolidate, analyze it and present users with the required insights – in real-time. These systems will also be capable of sending alerts to users, updating them with changes to the data at hand.

By becoming increasingly embedded in existing (and upcoming) workflows, these systems will allow teams of employees to carry out their day-to-day operations

and make decisions through the exploration of data in real-time. In addition to allowing for data analysis within existing systems, they will also integrate seamlessly with third-party systems – therefore, paving the way for an enterprise-wide, data-driven culture.

Lastly: Tackle the Latest BI Trends Head-On

Companies who want to stay on the right side of digital disruption must move fast and get serious about their data and analytics efforts. While much of the data analysis going on today is based on historical data, the focus is rapidly shifting towards a more forward-looking (and even automated) data-driven decision-making. BI landscape is soon going to be defined by trends around data governance, self-service BI, prescriptive analytics, NLP, BI-as-a-Service, and collaborative and integrative BI. If organizations want to stand a chance, they must embrace these trends and strengthen their foothold in today's competitive and dynamic business environment.

Augmented Analytics

Augmented analytics is an approach to data analytics that employs advanced technologies such as AI and machine learning to automate data preparation, insight generation, and insight sharing.

A primary distinction between augmented analytics and traditional analytics methods lies in the former's emphasis on automation and accessibility. Traditional analytics typically involve manual processes and require specialized skills in areas such as data science and statistics. These prerequisites can lead to bottlenecks in the analysis process, particularly when dealing with large volumes of data.

In contrast, augmented analytics automates many of these processes, making data analysis faster and more efficient. Additionally, it removes the barrier of technical expertise, allowing individuals without extensive data science backgrounds to understand and utilize complex datasets.

Data Storytelling

The growing dependence on data brings forth the need for data interpretation that extends beyond traditional methods.

The growing dependence on data in the corporate landscape brings forth the need for data interpretation that extends beyond traditional methods. A narrative layer aims to provide context and interpretation.

One of the primary differentiators between data storytelling and data visualization lies in the narrative structure. While data visualization can provide a

visual representation of what the data is saying, data storytelling goes a step further by explaining why the data matters, providing a more comprehensive understanding of the insights.

Data storytelling involves setting the scene and providing background information:

- The 'characters' in these stories are the various data points or metrics being discussed.
- The 'plot' usually involves a problem or challenge that the data can help address or an opportunity it reveals.
- The 'resolution' or conclusion of the story provides insights derived from the data, explaining what actions should be taken based on these insights.

Overall, data storytelling can illuminate complex market dynamics, customer behaviors, and campaign performance in an easily digestible format, enabling a more thorough understanding of the market landscape. Moreover, data storytelling can facilitate the translation of these insights into concrete, actionable strategies.

Decision intelligence

Decision intelligence (DI) is defined by Gartner as *"...a practical discipline that advances decision making by explicitly understanding and engineering how decisions are made and how outcomes are evaluated, managed and improved via feedback."* As an emerging trend within business intelligence, its focus is the use of AI technology to enhance, accelerate, and even automate decisions deriving from BI data.

DI software uses machine learning (ML) algorithms to process a company's big data and creates a semantic layer of business rules and contexts to ground predictive analytics which produce forward-looking insights to improve an organization's decision making capabilities. Reports suggest that the global DI market is set to grow from \$13.3 billion in 2024 to \$50.1 billion in 2030.

Mobile BI

In 2023, it was estimated that almost 96% of the global digital population used a mobile device to connect to the internet. Given the ubiquity of cell phones and the growth of BYOD (bring your own device) in workplaces, it's little surprise that BI solutions optimized for mobile devices are set to become more prevalent. One

study, for example, forecast that between 2021 and 2026 the mobile business intelligence market will experience a CAGR (compound annual growth rate) of 22.43%.

Mobile BI provides business leaders with the convenience of on-the-go insights to support data driven decisions, wherever they are and whenever they need the information. The growth in cloud BI data sets and faster mobile internet speeds, has made interactive, mobile-optimized business intelligence an attractive reality.

Collaborative BI

Collaborative business intelligence platforms combine BI tools with common collaboration technologies – one example of many is Power BI with Microsoft Teams – to facilitate easy sharing of insights and strategic collaboration between different business functions. Collaborative BI platforms enable different teams to work together concurrently on data analysis, often integrating discussion threads into BI reporting in order to promote mutual problem solving, data driven collaborations and shared understanding.

Data quality management

A BI solution is wholly reliant on its data quality to provide accurate insight for informed decisions. Consequently, data quality management is an increasingly important component of business intelligence deployments. In fact, the market size for data quality tools is projected to grow to \$8.49 billion by 2030 from \$3.23 billion in 2023.

Business intelligence tools fuelled by poor quality data can turn would-be competitive advantage into actual commercial disaster, because:

- Financial investments in BI solutions are wasted
- Bad data creates bad data analytics which fuels bad decisions, corrupting business strategies
- Training generative AI applications using poor data undermines their effectiveness and decision intelligence

Data management solutions help cleanse, standardize, verify and align data sets, as well as validating information provenance. They help ensure the data in data-driven decisions provides a solid foundation for business strategies.

Process Intelligence

A recent study from HFS research showed that 88% of enterprise leaders expect increases in Process Intelligence investments. Business intelligence and Process Intelligence can be simultaneously considered parallel, complementary and (more recently) intersecting practices. Process Intelligence provides BI professionals with the understanding and means to action business intelligence insights – providing the ‘how’, ‘why’, and ‘where’ processes need to be optimized in order to achieve business goals.

Process Intelligence, provided through platforms like the Celonis, enriches an organization’s business intelligence foundation with process insights, enabling business leaders to identify value opportunities in the data, pinpoint the processes to enhance, and respond quickly and effectively.

Who is an Information Worker?

When Peter Drucker originally articulated the idea of a “knowledge worker” in 1959, he was proposing a classification with the primary goal of describing the work of people who applied knowledge directly and in a unique way, to the tasks assigned to them. One important differentiator of the knowledge worker was that he or she owned their means of production—which literally means people have brains, and they bring their brains to work. Unlike blue-collar workers who do not own the factory equipment they use to produce products, knowledge workers own the knowledge and skills they apply to create value. Some use the term knowledge worker as a synonym for professional, which now accounts for roughly twenty-five percent of the workforce in industries like financial services, high-tech, healthcare, pharmaceuticals, and media. Knowledge work, in this now-classic definition, is too narrow to define the type of work that have evolved over the last thirty years.

As computing technology infused into organizations, it became a new tool for understanding an organization’s data. In many cases, it did little more than consolidate data, and make the data more presentable to knowledge workers. Over time, though, computer programming became more sophisticated. Some of the knowledge of knowledge workers found its way into computer programs and computers became more capable of applying knowledge to data without human intervention. This development started to spread knowledge through the organization, but usually in very limited, and very specific ways. This was the first

instance of information work: people without fundamental knowledge were able to apply the knowledge of others to data, and act on those insights without a direct connection or consultation with the holder of the knowledge.

An Information Worker is a person who uses information to assist in making decisions or taking actions or a person who creates information that informs the decisions or actions of others.

The application of computing power to data was also becoming more accessible, as programs developed that allowed non-computer specialists, usually knowledge workers, to create documents and databases that codified their knowledge. As these applications were shared, they too added to the disbursement of knowledge throughout organizations, but specialized knowledge remained far from pervasive. Knowledge work had expanded from those who knew personally, to those who could leverage knowledge embedded in computer systems.

As the personal computing revolution took hold, it became apparent that work itself was changing. The percentage of people working with data was increasing, and was no longer limited to the creation, collection, and forwarding of that information to “knowledge workers.” Data was staying local and being used to help workers of all types make better decisions about their work product, in real-time. These workers did not have throngs of knowledge workers to send their work to for interpretation and augmentation but rather used the knowledge that was now embedded in software to filter, organize and garner insight from the data most relevant to them. These workers became known as information workers. An information worker, therefore, is a worker who uses information to assist in making decisions, taking actions or creating information that informs the decisions or actions of others.

Let us step back, however, and use manufacturing to explore the evolution of information work and its relationship to business operations. It can be argued that blue-collar workers applied knowledge all the time as well. They just did so in ways that increased the value of physical assets rather than information or financial assets.

A knowledge worker in a front office would be hard pressed to understand the kind of work that takes place in a sheet metal factory, or the knowledge required to transform sheet metal into products. The front office worker may have clear visibility into the schedule, the costs, the status and other information elements of the factory, but they have no specific knowledge of the work being done.

If you think about a sheet metal worker, they have very deep knowledge of the material they work, and the machines used to shape the material into products

or components of products. A knowledge worker in a front office would be hard-pressed to understand the kind of work that takes place in a sheet metal factory, or the knowledge required to transform sheet metal into products. The front office worker may have clear visibility into the schedule, the costs, the status, and other information elements of the factory, but they have no specific knowledge of the work being done. This represents a two-tiered, hierarchical information model for the factory. One layer is specific and local, in that the manager and the sheet metal workers understand the knowledge required to transform raw materials into products and components. The other layer is abstract and represented in a language knowledge workers use to understand the factory. They treat the shop floor not as a physical location, but as a component of the business's information model. In this example both of the groups, those on the factory floor applying knowledge to material, and those in the factory's front office applying knowledge to schedule and cost control, are knowledge workers, and the interplay between their information makes them information workers in general.

Knowledge is specific, and information is general. Both of them use the same information about dates, schedules, plans, and costs. Updates to status come from the factory floor to planners in the front office who can anticipate material shortages, or reschedule the factory based on new demand. And the manager of the sheet metal shop reacts to the information coming from planners to create his or her daily schedule. That is information work. The knowledge work of both remain as well, with the planners applying knowledge to optimize factory operations in total, and the manager applying knowledge to make sure products are produced with high quality, high-efficiency in a very safe and well-maintained environment. Those specific examples of knowledge would rarely be exchanged between the parties, whereas the information about scheduling, status, and cost, flows freely between the two, even though it may be applied in different ways, with different levels of abstraction.

At a higher level, current efforts to distribute work in a way that reflects economies of scale, or to leverage lower-cost facilities and labour are the result of knowledge work applied at the abstract level of the information model. The actual work still requires local knowledge. That manufacturing knowledge remains consistent and local if it is applied in Detroit or in Bangkok. The more abstract information treats both locations as part of an overall business model that represents global production goals, including cost, transportation to various markets, and other factors. The differentiation between knowledge work and information work

remains consistent even when the definitions are applied to modern global enterprises.

Like any classification, information work begs for a more granular taxonomy, but it is at the highest level of abstraction that the profound implications become apparent. Information workers now have access to vastly more data and information than even the most astute knowledge worker had in 1959, and they have become better consumers of this information. This has precipitated a more localized and distributed view of decision-making, which has led to organizations that act more like networks, than the previous model that relied on hierarchies fashioned on a command-and-control model.

In fact, it can be argued, that knowledge work as a job classification was a reaction to the increasing amounts of data available in organizations, and the lack of a means to process that information. Information work focuses on a distributed model for empowerment and trust that sees the organization as a self-sustaining network with information feedback loops that reinforce positive, goal-oriented behavior.

Information work will become the predominate attribute of most occupations as work is transformed by the availability of data, and the ability to better understand that data so that local decisions can be better informed. The factory manager has more information about the factory, on a number of dimensions, from quality, to supplier relationships than ever before. And front office workers have more information about the overall business. This democratization of information has led to networked organizations where decision-making is more distributed, and where “management” often moves from day-to-day concerns around operational details to a role where they ask deeper questions about direction, vision, and values.

Top 7 BI Implementation Challenges to Consider

1. Ineffective Data Architecture

One of the key challenges in developing business intelligence solutions is designing data architecture. Basically, it forms the foundation upon which the entire BI ecosystem rests and impacts the effectiveness, scalability, and sustainability of BI solutions.

A well-structured data architecture ensures that information is organized, integrated, and maintained in a consistent manner. On top of that, it streamlines data access, eliminates silos, and provides a single source of truth across users. However, there is a great variety of information sources, including structured, semi-structured, and unstructured, so integrating and managing diverse data types becomes a daunting task. To keep everyone in the company interacting with the same reliable and up-to-date information, your team has to build a flexible and agile data architecture.

2. Poor Data Quality

Next on our list of BI challenges is data quality. The accuracy of business intelligence solutions relies on the data they are built on. Even with substantial investments, dedicated efforts, and a skilled team in place, you can't succeed if you use incomplete, missing, or outdated information.

To harness such kinds of business intelligence problems, data management can serve as a helping hand. It includes such practices as profiling, real-time monitoring, and regular information cleansing. At its core, data management ensures that information is secure, accessible, and correct.

Discover How to Achieve Better Data Management

To ensure data quality (DQ), you can define key data quality metrics such as accuracy, completeness, consistency, timeliness, or uniqueness. By establishing clear and measurable criteria, it is possible to both benchmark relevant information and identify areas that need improvement.

Additionally, there are many methods to monitor data quality, such as creating reports for tracking DQ or incorporating a DQ indicator into your current reports. These techniques allow you to determine how reliable and accurate the data you receive.

3. Unclear BI Strategy

Another one of the significant challenges of BI solutions is the lack of a clearly defined business intelligence strategy. And really, how can you guide the implementation process and ensure that the organization goals and objectives are met effectively without an appropriate plan?

If you don't have a clear strategy, the scope of your BI projects can expand beyond the intended. As a result, you can face delays, increased costs, and difficulties in getting meaningful insights.

Without a solid plan, you might also end up with BI tools that don't meet the needs of end users. This can cause low user adoption rates and frustration among employees who struggle to find value in the BI tools.

As you can see, underestimating the significance of a well-implemented business intelligence roadmap can lead to a range of additional problems. So take your time to carefully plan and execute your data-driven strategy.

4. Weak Content Management

Companies that overlook the importance of effective content management can face many challenges during the business intelligence implementation process. As a quick side note, here content management means handling and organizing the data itself, reports, dashboards, and other information generated by BI systems.

Obviously, employees are more likely to embrace BI solutions when they can easily find and use the content they need. Appropriate content management facilitates quick search and retrieval of specific reports or data.

It's also important to create documentation for each of these reports since it provides clear explanations of the metrics used to assess data. Documentation ensures users understand and appropriately interpret data to make better-informed decisions.

Additionally, it is worth notifying your employees about new reports or updates to existing ones. Thus you can help your team to keep up with the latest insights. On top of that, alongside effective content management, it is critical to deliver information in an accessible and simple manner. With that in mind, consider designing user-friendly interfaces for BI tools. Such UIs can present complex data in a simple manner utilizing charts, graphs, and other visualization.

Without a well-crafted design, users may be unwilling to engage with BI solutions. This, in turn, can significantly contribute to lower rates of BI adoption. More on that we will talk in the upcoming chapter.

5. Low Adoption Rates

One of the common problems that can arise during business intelligence implementation is low adoption rates. You may face challenges in encouraging your staff to employ business intelligence tools. For example, they may be reluctant to say goodbye to Excel, SaaS platforms, or other apps they are so comfortable using.

Being unfamiliar with BI tools is just one reason why people do not want to use them. Another reason is a deficiency in data literacy. If you don't have skilled employees who can interpret data accurately, then possibly you can't make informed decisions. Whereas, data-literate employees can assess the credibility of data sources, understand the context, and draw meaningful conclusions.

Consequently, if you want to harness the full potential of your business intelligence solutions and tackle problems on time, then it is definitely worth fostering data literacy among your team.

To get to the point, you may offer training or master classes to your staff. Also, consider opening support channels in your team's message board to provide timely responses to your employees queries.

As a result, you can build a community of data-literate individuals within your organization who can help others and share their expertise.

6. High Fees

We've almost come toward the conclusion of the list of business intelligence challenges and we can't omit the substantial expenses it requires. Pricing for BI software might vary depending on:

- data volume
- company size
- project dimension
- customization
- number of users
- complexity of the implementation process

Typically, you need to consider costs for licenses, hardware specifications, employee training, and software maintenance. The latter can come with additional fees.

That is to say, implementing BI is not a one-time endeavor, but rather an ongoing process that demands constant development and investment. Business intelligence tools need to evolve alongside a company's changing needs and its existing data landscape.

So, as a garden requires constant care to thrive, a successful BI implementation requires sustained attention, which may cause additional expenses. Yet, it is definitely the case when such expenses are reasonable.

Companies that opt to cut corners in efficiently processing data run the risk of being pushed out of the competition. Hence, investing in robust analytics and BI tools is a pathway to success.

7. Hard to Deliver Mobile BI

The list of business intelligence challenges will be incomplete without mobile BI. Though adapting business intelligence tools to mobile devices appears to be simple, it doesn't mean that this process is devoid of complexities.

Here are some basic issues you can come across along your way:

- **Data Security.** Mobile devices can be more susceptible to security breaches, so employing strong encryption and authentication measures is crucial.
- **User Experience.** With different screen sizes and a variety of device types, it's vital to think about how users can smoothly navigate and interact with your BI solution on their mobile devices.
- **Performance and Speed.** Mobile networks might not always provide the same level of connectivity and speed as traditional networks.

Overall, to successfully utilize mobile BI, you need careful planning, design, development, testing, and ongoing maintenance. On top of that, your team should be experienced in BI, web, and mobile technologies.

According to a recent study, the adoption rate of global business intelligence solutions stands at 26%, with this figure tending to rise in larger organizations. Despite this significant adoption rate, the majority of organizations are likely to encounter business intelligence challenges during the implementation phase.

It is essential for all organizations to strategically plan and address these barriers in order to overcome them successfully. This includes implementing BI best practices and staying updated with the latest developments and topics in the field to fully leverage the associated benefits.

The elements of business intelligence are a critical factor of every organization's success. Businesses will be unable to get insights, respond to changes, manage the BI cycle, or make educated business decisions unless they have the ability to easily analyze data.

The business intelligence challenges are:

1- Data Integration from Different Source Systems

BI is only effective when it can consolidate and visualize data from all important repositories that exist in your organization to provide the end user the ability to

gain holistic insights. If that is not achievable, there is no point to invest and try to implement BI in any organization.

Businesses should make sure that all data repos including databases, spreadsheets, cloud applications, social media, IoT devices, and ETLs available that contain important information to digest and analyze should be able to integrate with the adopted tools in order to allow senior management to have a complete overview of the data and be able to make better decisions based on full data and not a subset.

This represents a significant obstacle that will negatively impact the implementation and the quality of the decisions being made. That is why organizations are required to carefully plan and overcome it.

2- Data Quality Issues

Data quality is another significant challenge in business intelligence as it represents the core element to getting the most out of this initiative. With poor data quality, analyzing, slicing, and dicing data will jeopardize the ability to effectively gain insights and understand how the business operation is performing.

Organizations should make sure their data management strategy has clearly defined how to improve their data quality by ensuring the accuracy, consistency, and completeness of data.

Data entry errors, incomplete or missing data, outdated information are major factors of having poor data standards in your organizations. This will lead to unreliable analysis, flaws in predictions, and inability to identify patterns.

What we recommend usually is to prioritize having high standard quality of data by automating data entry, implementation data validation processes, and data cleansing techniques.

3- Complex Analytics

Advanced analytics methods like machine learning and predictive modeling can offer insightful data. However, there are difficulties in terms of talent, infrastructure, and budget allocation when adopting and integrating these advanced analytical capabilities.

The difficulties rely on having very well trained and skillful resources that are familiar with these advanced techniques to drive the best analytical experience throughout your organization. Also, you need to make sure to improve their employment retention to keep them working within your network as long as possible.

4- Low User Adoption

In certain instances, despite the setup of the software appearing to have gone smoothly, the project management being happy that the solution can accomplish its goals and users having access, the solution is not being used.

Throughout my experience with digital solutions implementation, I came to a conclusion that when the user adoption is low that usually means the software doesn't provide an intuitive user experience that makes it easy for employees to work with.

A bad user experience is often accompanied by extended training periods, inefficiencies in task completion, and a direct impact on overall productivity. Moreover, it can be a crucial factor in the loss of invested money.

5- Ineffective Data Visualization and Dashboards

Another critical challenge that organizations face when implementing business intelligence is having ineffective data visualization and badly designed dashboards that can't deliver what the senior managers are looking for.

Having an effective BI implementation requires to have all the pieces of the puzzle correctly placed. If data is of high standards and all other factors are available but the way you deliver the message by designing charts and reports is inaccurate that means you are bound to fail as it won't help your company as intended.

Possible problems with integrating the dashboard include a lack of interaction, the inability to pull data in almost real-time, rigid layouts, and even the incorrect use of color.

6- Creating a Data-Driven Culture

Establishing a data-driven culture in any organization is one of the most critical business intelligence challenges because it requires everybody on board to have a fundamental shift in their mindset and daily operations. Many struggle to achieve this because of many reasons including resistance to change, lack of the power of data, and inability to cope with the technological changes.

To overcome this barrier, organizations should be providing continuous training and support for employees that feel uncomfortable, promoting transparency, and highlighting the value of data in this digital world.

7- Managing the Use of Self-service BI

To efficiently harness the power of BI, organizations should tend to give their employees the freedom to access, evaluate, and analyze data on their own without the need for IT interference.

Without relying on IT, data scientists, or analysts, self-service BI enables business users to examine data, draw insights, and build dashboards or reports. Self-service business intelligence is based on the idea that all employees, regardless of analytics expertise, should have access to information that will help them make better decisions.

8- Lack of Skilled Personnel

The role of business intelligence relies heavily on professionals who can interpret and analyze complex data sets, design meaningful visualizations, and implement robust BI strategies.

Unfortunately, the demand for these skills often outpaces the supply, leading to a deficiency of qualified BI professionals. This shortage can impact an organization's ability to fully leverage the potential of its data for informed decision-making.

Addressing this challenge requires investments in training programs, education initiatives, and talent acquisition to ensure that organizations have the skilled workforce needed to navigate the complexities of modern workspaces.

9- Data Silos

Data silos occur when information is isolated within specific departments or systems, hindering the seamless flow and sharing of data across an organization. This fragmentation can impact the effectiveness of BI efforts as a comprehensive and unified view of the data becomes challenging to achieve.